Does Greener Demand Induce Cleaner Production? Green Procurement Competition and Fuel Switching in the U.S. Cement Industry

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Abstract

As carbon pricing faces political barriers, governments and large private buyers increasingly turn to green procurement policies, buying only from low-carbon suppliers to encourage clean production. This paper studies how Buy Clean initiatives reshape competition and clean investment in the U.S. cement industry. I develop a two-stage structural model in which plants choose whether to switch from coal to natural gas and then compete in procurement auctions. Counterfactual simulations show that Buy Clean induces plants to switch to cleaner production but also exacerbates market power distortions. Restricting public purchases to low-carbon suppliers weakens competition, increasing their markups on green sales. Shifting demand toward clean plants pushes their output closer to productive capacity, raising their marginal costs and increasing their bids to all buyers. Dirty plants' markups also rise as they face less competition from clean rivals. A border-adjusted carbon tax that induces the same switching rate cuts more emissions at a lower welfare cost per ton of CO2 avoided.

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1 Introduction

Green procurement has increasingly emerged as a climate policy instrument for governments and large private buyers, as carbon taxes and cap-and-trade, long favored by economists, face persistent political and institutional barriers in the United States (Krupnick 2020). Under a green procurement standard, a buyer conditions its purchases on products' carbon intensity. Clean suppliers below the carbon limit gain eligibility for public contracts, and dirty suppliers above it lose it. The policy uses large purchasing power to make a sizable portion of market demand favor green products and thereby reward cleaner production. As the world's largest buyer, US governments at the federal and state level have applied such criteria to public contracts across products, from energy-efficient appliances to renewable electricity [1]. Private buyers, including technology companies building data centers, increasingly specify low-carbon construction materials to meet their own carbon targets [2] Green procurement is therefore a demand-side push for clean investment, yet little is known about how it induces that investment or how it compares with carbon pricing in effects and efficiency.

Green procurement can reshape market competition: the eligibility rule allows only the cleaner suppliers to compete for a green buyer's contract, and in a concentrated market a clean supplier can charge a weakly higher markup and price as it faces weaker competition from clean rivals. The policy may also affect the dirty suppliers, and through them the brown buyers who do not purchase on carbon performance. A plant's gain from becoming eligible therefore depends on its cost structure and carbon intensity, on which rivals are excluded, on how prices change for the brown buyers the rule does not cover, and on the investment cost, so investment and competition must be modeled jointly. While most polluting industries are highly concentrated (Fowlie et al., 2016; Preonas, 2024), little work studies how an eligibility-based climate policy interacts with existing market power.

To study how green procurement reshapes investment and competition, I develop a two-stage structural model of fuel switching and procurement competition in the U.S. cement industry and use it to simulate the effects of the Buy Clean procurement policy. In the model, a plant's investment in fuel switching changes its marginal cost and its carbon intensity, and therefore how it competes for procurement contracts against nearby rivals. Under a green

[1] Federal agencies are required to purchase ENERGY STAR certified products (EPA, "Federal Procurement Policies for ENERGY STAR Certified Products," https://www.energystar.gov/products/federal_procurement_policies_energy_star_certified_products) and procure green power (EPA, "Green Power Procurement Considerations," https://www.epa.gov/green-power-markets/green-power-procurement-considerations)

[2] Microsoft, "Low-Carbon Building Materials for Data Centers," https://news.microsoft.com/source/features/sustainability/low-carbon-building-materials-for-datacenters/.

procurement standard, the fuel choice also determines plants' eligibility for green contracts, so the return to switching includes the gain from the competition the policy restricts. This framework allows me to examine whether that return is large enough to cover the switching cost. With the estimated model, I simulate Buy Clean designs alongside a border-adjusted carbon tax and compare their effects on switching, prices, markups, emissions, and welfare.

The cement industry is carbon-intensive and locally concentrated, and switching the production fuel from coal to natural gas is a practical near-term abatement option that can move a plant into policy eligibility. Cement accounts for about 8 percent of global CO_2 emissions, and the switch cuts combustion emissions by roughly 24% (Portland Cement Association, 2024). Because cement is bulky and costly to ship, few suppliers compete for any given buyer, and plants hold substantial local market power (Ryan, 2012; Miller and Osborne, 2014). The switch is a large, irreversible investment, and adoption has been slow and uneven, shaped by fuel prices and pipeline access: by 2022, about 40% of U.S. plants used natural gas as a primary fuel, while the rest remained on coal or petroleum coke.

The Buy Clean Initiative is the first federal green procurement policy targeting the embodied carbon of construction materials, with eligibility set by a maximum allowable carbon intensity for the product (U.S. General Services Administration, 2023). Suppliers must publish an Environmental Product Declaration (EPD) reporting the product's global warming potential (GWP), and a supplier is eligible when its GWP is below the threshold. Where earlier work measures energy efficiency from kiln technology alone (Ryan, 2012; Fowlie et al., 2016; Miller and Osborne, 2021), GWP gives plant-level carbon intensity, so eligibility is empirically traceable to each plant's fuel mix and technology. With roughly half of U.S. cement going to public construction, the standard can shift producer incentives at scale. The federal initiative was canceled in 2025, but its partnerships with 13 states continue, and most state standards are only beginning to take effect in 2026.

The two-stage model links the fuel switching with procurement competition. In the first stage, plants make a static investment to switch from coal to natural gas, weighing the switching gain in procurement profit against a switching cost that depends on the distance to the pipeline network, its prior experience with gas, and the air pollution compliance cost it avoids (Carlson et al., 2000; Scott, 2023). In the second stage, plants compete for procurement contracts in spatially concentrated markets through a second-score auction building on Miller et al. (2025). Each plant bids at its marginal cost, which rises with fuel prices and capacity utilization, and the transaction price is set by the second-best competitor's score.

To estimate the model, I compile a plant-level dataset on fuel use, kiln technology, and

production capacity, combined with regional quantities and average prices data and transportation networks. For the second stage, I estimate a nested logit demand model for county buyers weighing suppliers' marginal cost, distance-based transportation cost, and preference for imported cement, using variation in fuel costs, capacity, and plants' location (Miller et al., 2025; Ellickson et al., 2020). The results show that transport costs are central, fuel costs pass through to bids, and marginal cost rises steeply as plants approach capacity. The first stage recovers the switching cost from observed switching decisions following revealed preference logic with the moment inequalities of Dickstein and Morales (2018) and Pakes et al. (2015). The cost is $67 to $82 million per plant in present value, several times the engineering cost of the pipeline connection; prior gas use lowers it, and distance to the pipeline raises it. Held out of estimation, the model predicts the switching observed between 2019 and 2022, validating the estimated market structure.

To simulate the Buy Clean as a counterfactual, I introduce heterogeneous buyers and model the maximum allowable carbon as an eligibility constraint to the supplier set. Green buyers, including public buyers subject to Buy Clean standards and private buyers committed to low-carbon construction, incorporate the carbon performance into their procurement criteria. They only purchase from the eligible plants whose carbon intensity is below the policy threshold. Brown buyers do not impose such constraints and procure as conventionally. I simulate different Buy Clean designs, varying state or national policy coverage, the green-buyer share, and the carbon threshold, with a border-adjusted carbon tax.

I find that Buy Clean induces switching, cuts emissions, and exacerbates the existing market power. Under the current design in the 13 partnership states, the policy induces around 20% of domestic capacity to switch to gas and cuts emissions by 2.5%. A national Buy Clean induces close to half of domestic capacity to switch and cuts emissions by 6.5%. Green buyers move their purchases from dirty to clean plants, so clean plants run closer to capacity and their marginal cost rises, while dirty plants lose volume and their marginal cost falls. Clean plants charge a higher markup and price to green buyers, since the dirty plants can no longer bid for them and the second-best bid is weakly higher. In the brown market, the clean plants bid at their higher cost and become less competitive, so dirty plants also charge a higher markup, and prices rise for conventional buyers as well.

The state-level Buy Clean, as a regional policy, creates a policy spillover and moderates the market distortion. 80% of the dirty output displaced from green buyers reappears in non-Buy Clean counties within a few hundred miles. The dirty plants keep a market to sell to, so they keep bidding against the clean plants and their markup in the brown market rises by only 1%. Emissions still fall in the non-Buy Clean counties, because the plants that

switch serve those markets: 30% of the emission cut occurs there. A national Buy Clean leaves no non-Buy Clean counties to absorb the displaced output. Dirty plants shed volume, and clean plants bid higher to everyone. Dirty plants' markup in the brown market rises by 6%, and brown buyers face a price increase close to that of green buyers.

The welfare cost of Buy Clean is $195 million a year under the 13-state design and $437 million under the national design, or $88 and $78 per ton of CO2 avoided, below the EPA social cost of carbon of $190. While green public buyers lose $215 million, brown buyers also lose $54 million in the state design because of the market distortion. Most of it goes to the eligible plants as scarcity rents, and the plants that switch gain $58 million net of their conversion cost. A border-adjusted carbon tax of $50–75 per ton induces the same switching share as the state-level Buy Clean, cuts emissions by 11–17%, and costs $19–32 per ton of CO2, because it charges every ton of emissions and what buyers pay returns as public revenue rather than remaining with the eligible plants.

This paper contributes to three strands of literature. First, this contributes to the literature in green procurement and demand-side climate policy. Past empirical work discusses how policies affect firm behavior in technology adoption and innovation (Krieger and Zipperer, 2022; Zhang and Jiang, 2022; Zhang and Zhu, 2024; Zheng and Wen, 2024) through information and signaling channels (Simcoe and Toffel, 2014; Matisoff et al., 2016), and financial channels through which procurement contracts reduce revenue uncertainty (Liu et al., 2025). I contribute to this literature by introducing heterogeneous buyers and modeling policy thresholds as supplier eligibility rules. This allows discrete choice modeling to capture buyers' preferences in carbon intensity when buyers' continuous preference for environmental quality is not observed. I then provide policy implications to explain green procurement as a less efficient but politically easier instrument, while carbon tax is more efficient but difficult to implement.

Second, I contribute to the literature on how market power interacts with environmental policies. Past literature has found that environmental regulation exacerbates existing market power distortions in concentrated markets (Ryan, 2012; Fowlie et al., 2016), and market power can change the efficacy of environmental policies (Mansur, 2013; Leslie, 2018; Preonas, 2024). This paper is closely related to the following paper. Drake et al. (2024), who find that stricter environmental criteria in procurement reduce entry and raise prices through administrative complexity. Aghion et al. (2023) use country-level data to study how green preferences drive clean innovation in the automobile industry, with country-level competition held fixed. I link to the literature by presenting how green procurement eligibility rules exacerbate market power by segmenting the market and changing the supplier set. This

policy design mechanically reshapes the market structure, and therefore exacerbates the existing market power.

This paper also contributes to the literature on how environmental regulation shapes firms' investment in cleaner production technologies. Existing work shows that regulation affects abatement technology adoption through fuel prices, compliance costs, and regulation design in electricity (Fowlie, 2010; Fowlie et al., 2012; Fell and Linn, 2013; Carlson et al., 2000; Scott, 2023) and manufacturing (Anderson and Newell, 2004; Linn, 2008; Shapiro and Walker, 2018). In cement literature, prior work examines emissions regulation (Fowlie et al., 2016; Ryan, 2012) and technology adoption (Macher et al., 2021; Miller and Osborne, 2014; Miller et al., 2025), but little work has examined fuel-switching investment. This paper relates to earlier conceptual work: Amacher, Koskela, and Ollikainen (2004) propose a model of vertical product differentiation linking investment, environmental quality provision, and price competition, and argue that firms' incentives to invest depend on relative cost structure. I build on this literature to study fuel switching investment using a framework studying policies linking endogenous investment in carbon performance to downstream market procurement competition.

The rest of the paper is organized as follows. In Section 2, I describe the policy context and data, and present key stylized facts that motivate the subsequent structural analysis. In Section 3, I introduce my two-stage structural model. In Section 4, I describe the estimation strategy. After I present my estimation results in Section 5, I use Section 6 to study the counterfactual exercise of Buy Clean policies. I conclude this paper with Section 7.

2 Policy Background and Data

2.1 Green Procurement: The Buy Clean Initiative

The United States federal government is the largest purchaser of goods and services in the world, with annual procurement exceeding $630 billion. Public procurement has long been used to promote broader policy objectives by favoring certain types of firms or products, such as minority- and women-owned businesses, small businesses, or environmentally preferable goods. Because government contracts provide large-scale, stable demand and predictable revenue streams, procurement policy can substantially reshape market outcomes and create incentives for firms to innovate or adjust production decisions. In recent years, federal procurement has become an increasingly important instrument of U.S. climate policy, particularly through efforts to reduce emissions embodied in construction materials. Since roughly

half of U.S. cement demand is tied to public construction and infrastructure projects, these procurement policies have the potential to alter cement producers' incentives at scale.

The 2021 Federal Sustainability Plan established a target of net-zero emissions from federal operations and procurement by 2050, with particular emphasis on emissions embodied in industrial supply chains. Within this broader framework, the Federal Buy Clean Initiative, launched in 2021, specifically targets embodied emissions in construction materials such as steel, cement, concrete, and glass. These industries are among the most emissions-intensive sectors of the economy due to energy-intensive production processes and limited short-run substitution possibilities.

Buy Clean policies prioritize the procurement of construction materials with lower embodied carbon in federally funded projects. The policy relies on product-specific Environmental Product Declarations (EPDs), which provide third-party verified life-cycle assessments of environmental impacts, including global warming potential (GWP) and other pollutant emissions. Suppliers are required to disclose EPDs for eligible products, allowing public agencies to compare embodied emissions across producers. Public agencies then establish maximum acceptable embodied-carbon thresholds, typically measured using GWP benchmarks. Construction materials that satisfy these thresholds qualify for procurement under Buy Clean standards.

The federal initiative has been complemented by the Federal-State Buy Clean Partnership, launched in 2023, which includes states such as California, New York, and Maryland. Participating states commit to prioritizing low-carbon construction materials in public projects and to coordinating procurement standards across jurisdictions. At the same time, private-sector commitments to the use of low-carbon construction materials have also increased, particularly among large technology companies facing rapid expansion in data-center construction. These firms increasingly incorporate embodied-carbon considerations into procurement decisions and have supported the development of lower-carbon cement and concrete technologies.

The Inflation Reduction Act initially allocated approximately $4.5 billion to support the procurement of low-embodied-carbon construction materials. Although portions of the federal program have since been terminated, state-level Buy Clean programs and industry initiatives remain active, and public agencies and industry associations continue to develop embodied-carbon standards for construction materials.

2.2 Cement Industry

Cement is a fundamental input in construction and infrastructure, serving as the key binding material in concrete used in buildings, roads, bridges, and public works. Cement production is highly energy- and carbon-intensive. The manufacturing process involves heating limestone and other raw materials in large rotary kilns at temperatures exceeding 1400°C to produce clinker, the primary intermediate input in cement production. The clinker is then mixed with gypsum and ground into portland cement. During this process, emissions arise both from fuel combustion used to heat the kilns and from calcination, the chemical reaction that releases carbon dioxide from limestone.

Several technological pathways have been proposed to reduce emissions from cement production. These include kiln efficiency upgrades, fuel switching, clinker substitution through supplementary cementitious materials (SCMs), carbon capture and storage technologies, and electrified kiln technologies. Some approaches, such as carbon capture and electrified production, have the potential to reach zero emissions but remain costly and under technological development. Other approaches, such as kiln efficiency improvements through the transition from older wet-process kilns to more efficient dry-process kilns with preheaters and precalciners, have already been widely adopted. As a result, the remaining potential for emissions reductions through further kiln upgrades is relatively limited.

Among currently available abatement options, fuel switching remains one of the few ongoing near-term approaches based on mature and commercially available technology. Cement plants can shift away from carbon-intensive fuels such as coal and petroleum coke toward lower-carbon alternatives, such as natural gas and alternative fuels, while continuing to operate within existing production systems. Although fuel switching cannot eliminate process emissions from calcination, according to the Portland Cement Association (2024), replacing coal with natural gas can reduce combustion emissions by around 24% in the United States. Alternative fuels have also been increasingly adopted, primarily as secondary fuel inputs supplementing coal or natural gas. These fuels include waste-derived materials such as tires, biomass, and industrial byproducts. Their supply conditions are often shaped by local waste-management markets and transportation constraints. The use of alternative fuels is often more difficult to characterize within a unified national fuel market framework. In practice, switching from coal to natural gas may require new burner systems, pipeline interconnections, delivery modifications, and adjustments to kiln operations.

In cement, the effects of the Buy Clean policy can usually be indirect: agencies often buy ready-mix concrete rather than cement, but the concrete's environmental product declaration

embeds the upstream carbon profile of the cement used in the mix. That means procurement rules aimed at concrete GWPs can reallocate demand toward cement plants with lower carbon intensity.

2.3 Data

Plant Information Summary

The primary source of plant-level data is the Plant Information Summary (PIS), published by the Portland Cement Association. I use PIS observations for 2004, 2006, 2008, 2010, 2013, 2016, 2019, and 2022, yielding an unbalanced panel that spans two decades with varying frequency. Each observation records the end-of-year status of a U.S. cement plant, including its geographic location and ownership structure by company. At the kiln level, the PIS reports the kiln age, clinker production capacity, primary fuel type, operating status (active or idled), and manufacturing technology (wet process, dry process, or dry with precalciner). Capacity is expressed as an annual clinker tonnage that already incorporates a standard allowance for scheduled maintenance downtime, and therefore represents the maximum feasible production under normal operating conditions.

It is worth to first clarify the conceptual distinction between clinker and cement. Cement production proceeds in two stages: energy-intensive clinker production, which drives fuel demand and is capacity-constrained, followed by relatively low-cost blending and grinding. Market shipment data are measured in cement, whereas PIS capacity data are reported in clinker. To bridge these two units, I assume a fixed clinker-to-cement conversion ratio, following standard practice in the literature and PIS data [3].

USGS Minerals Yearbook

Market-level quantity and price data come from the Minerals Yearbook, published annually by the U.S. Geological Survey (USGS). The Yearbook aggregates plant-level reports to a regional geography. I use three tables from each annual volume. Cement consumption in each region is measured by shipments of Portland cement to final customers. These data capture deliveries across all varieties of Portland cement in both bulk and bag form. Regional supply is measured by Portland cement shipped from each region's plants. This quantity represents ex-plant sales, including shipments through a plant's own distribution terminals. The average regional price is the mill-net (ex-plant, free-on-board) valuation per metric ton, calculated as total revenue from sales to final customers divided by quantity shipped. Be-

[3] PIS started to report clinker-to-cement ratio since 2010. Data shows this ratio is consistently around 93%-89% before the adoption of Type IL around 2022.

cause the Yearbook also reports masonry cement, I restrict attention throughout to Portland cement, which accounts for roughly 90 to 95 percent of total cement consumption. Import quantities enter regional markets through U.S. customs districts (cities). USGS documents the imported quantity and customs value plus insurance, freight, and other delivery charges to the first port of entry at each city.

Data for Measuring Market Size

Defining the market size for demand estimation requires a measure of the industries that consume cement. I construct this measure from the BEA Input–Output Total Requirements Table (industry-by-commodity), which quantifies the total direct and indirect cement input requirements associated with one dollar of final demand in each downstream sector. I use the industry-by-industry total-requirements table as a robustness check. Sector-level value added at the four-digit NAICS level is drawn from the BEA for each two-digit sector. County-level employment data at the industry level comes from the Quarterly Census of Employment and Wages (QCEW) program under the Bureau of Labor Statistics to disaggregate the cement market demand into the county level. These data allow me to construct a measure of downstream market activity that varies across counties and years and is plausibly exogenous to cement supply conditions.

Shipment Costs and Transportation Networks

Matching plants to markets requires estimates of transportation costs by mode. Road distances between each plant-market pair are computed using the Open Source Routing Machine (OSRM), yielding routed (rather than straight-line) distances. Waterway distances are computed from the National Waterway Network Lines and Nodes, published by the Bureau of Transportation Statistics, using shortest-path algorithms on the navigable waterway graph. Cost rates that translate distances into dollar-per-ton shipment costs are calibrated to the Commodity Flow Survey (CFS), which reports shipment values, weights, routed distances, and modes for manufacturing commodities. I use the CFS data to estimate cost-per-ton-mile relationships separately for overland and water-based travel, and apply these to the routed distances to obtain plant-market-mode cost matrices.

Fuel Input Costs and Other Data

I obtain state-level industrial fuel prices, covering coal, natural gas, residual fuel oil, and petroleum coke, from the Energy Information Administration's State Energy Data System (SEDS) to compute plants' marginal cost.

Proximity to fuel supply infrastructure is captured through two additional sources. Coal

mine locations and annual active counts come from the Mine Data Retrieval System (MDRS) maintained by the U.S. Mine Safety and Health Administration (MSHA). Access to natural gas is proxied by each plant's distance to the nearest interstate transmission pipeline, computed from the EIA's National Gas Pipeline maps.

2.4 Descriptive Statistics: Evidence For Fuel Switching

Coal and petroleum coke have historically dominated as the primary energy sources. Over the past two decades, the cement industry has gradually but incompletely transitioned from coal to natural gas. By 2022, about 43% of plants use natural gas as their primary fuel. Figure 3 Panel A shows a clear transition into natural gas. The persistence of coal-fired plants suggests substantial unrealized potential for further fuel switching.

Two key factors help explain this trend. First, the expansion of natural gas use followed the shale gas boom, which significantly increased domestic supply and reduced natural gas prices. As shown in Figure 3 Panel B, natural gas prices declined substantially after the late 2000s. This price shift lowered the marginal cost of switching and made natural gas a more competitive alternative for cement producers.

Second, the decline of coal mines may play an important role, as shown in Figure 4. The average number of coal mines located near cement plants has fallen substantially over time across all distance thresholds (50, 100, and 200 miles). The average distance between cement plants and their nearest coal mines has increased. The combination of fewer nearby mines and longer shipping distances implies a structural decline in coal accessibility and potentially rising transportation costs, reinforcing the incentive for plants to adopt alternative fuels.

On the other hand, access to local natural gas infrastructure has also significantly influenced fuel-switching decisions. While the shale gas boom substantially reduced natural gas prices in the United States, not all cement plants were equally positioned to take advantage of these lower prices. Anecdotal evidence and research reports suggest that plants lacking direct access to natural gas infrastructure faced substantial pipeline expansion barriers and costs before natural gas could be adopted at scale [4]

The transition has also been spatially uneven. As the Figure 6 shows, coal-fired plants

[4]For example, conversion to natural gas in the Maryland Union Bridge plant would reduce 20% of its CO2 emissions. However, the plant is located far from existing natural gas pipelines and therefore faces substantial permitting barriers and pipeline construction costs, slowing the adoption of natural gas. In a recent study, Heidelberg Materials estimated that the pipeline extension cost would be approximately $50 million (Kennedy et al., 2022). Similarly, the cement plant in Maine faces significant barriers to a transition to natural gas due to local opposition to the pipeline expansion.

are concentrated in the Midwest and Great Lakes, and natural gas adoption is emerging primarily in Texas, California, the Pacific Northwest, and parts of the East Coast. Figure 7 provides a finer temporal view of this transition by plotting the cumulative share of switched plants within each state over the sample period. Three features stand out. First, most states transition in a single discrete period rather than gradually over time. Second, four larger states with more plants—Pennsylvania, Florida, Alabama, and Texas—exhibit two distinct adoption waves, reflecting heterogeneous plant characteristics within the state market. In Alabama and Texas, the two waves occur in consecutive observation periods. Third, there is substantial cross-state heterogeneity in the timing of switching. Pennsylvania was the earliest adopter, with its first wave occurring around 2010, potentially impacted by the early surge of Marcellus shale gas production (Hausman and Kellogg, 2015). In contrast, California and Arizona transitioned relatively late around 2019.

Figure 8 combines the spatial distribution of cement plant fuel use with regional variation in relative fuel prices, illustrating that changes in relative fuel prices over time aligned with the geographic expansion of natural gas adoption. Regions with lower coal-to-natural-gas price ratios, particularly Texas, Arizona, the Pacific Northwest, and the South of the East Coast, experienced more concentrated adoption of natural gas. In contrast, many Midwestern and Great Lakes states continued to exhibit higher coal competitiveness and remained coal-dependent throughout the sample period.

To study the factors affecting fuel choice, Table 5 presents reduced-form evidence from a simple logit regression examining the determinants of natural gas adoption. I found that plants located closer to natural gas pipeline infrastructure and in regions with more favorable natural gas prices are significantly more likely to adopt gas, while coal-intensive regions remain less likely to transition. This pattern is consistent with the descriptive evidence on increased gas adoption following the shale gas boom. The results also suggest that plant characteristics and technological constraints play an important role in limiting complete industry-wide adoption. Measures of coal market conditions include the coal price and access to coal infrastructure. A higher coal price increases the likelihood that coal will be substituted away as a primary fuel. Distance to coal mines enters positively in column (4), suggesting that as coal sources become more distant, plants are more likely to use gas. These results support the interpretation that higher relative local coal costs and declining coal accessibility contribute to the existing fuel switching.

These adoption patterns motivate the modeling choice for fuel-switching decisions. Two features of the observed transitions support a static investment framework. First, switching within most states is concentrated in a single observation period rather than spread grad-

ually over time. This pattern suggests that plants switch in response to state-level shocks, including shale gas access, local coal-to-gas price differentials, and NESHAP compliance pressure since 2010, which required compliance within three years. Second, the substantial cross-state heterogeneity in switching timing is consistent with observable variation in state-level energy market conditions. The remaining within-state heterogeneity observed in larger states (Pennsylvania, Florida, Alabama, Texas) is consistent with plant-level differences in switching costs driven by pipeline access, kiln technology, remaining asset age, and ownership transitions, all of which are included in the structural model as observable plant characteristics. Although dynamic strategic interactions and option-value considerations may exist in principle, the observed pattern of within-state synchronization combined with observable cross-state drivers is broadly consistent with a static investment choice in response to local industry shocks with heterogeneous plant-level switching costs. I therefore model the adoption decision as a static investment choice and incorporate option-value considerations as an extension in the counterfactual simulation section.

3 Model

I develop a two-stage model of investment and procurement competition to analyze how demand-side environmental policy, specifically the Buy Clean Initiative, shapes fuel-switching decisions among cement plants. In the first stage, each plant decides whether to invest in switching from coal/petroleum coke to natural gas. This decision affects the plant's marginal cost and embodied-carbon intensity, which jointly determine its competitiveness in procurement. In the second stage, plants compete for procurement contracts in downstream markets, where buyers select suppliers based on price, non-price attributes, and eligibility under carbon-intensity criteria. The key linkage between the two stages is that fuel choice affects not only costs but also the set of competitors it faces under different buyer types.

Let $\mathcal{J}$ denote a finite set of cement suppliers indexed by j, categorized into domestic plants and import terminals. Each supplier is owned by a company $c \in \mathcal{C}$, with $\mathcal{J}^c$ denoting the set of suppliers owned by company c. There is a continuum of buyers indexed by i with total mass $M > 0$. Assume each buyer procures from a single supplier in $\mathcal{J}$, or opts for an outside good (indexed $j = 0$) supplied at a price normalized to zero.

3.1 First Stage: Fuel-Switching Investment

In the first stage, each plant j chooses whether to switch from coal, petroleum coke, or oil to natural gas as its primary fuel,[5] anticipating the procurement competition that follows. The decision is a one-time, static investment with a sunk cost (De Jonghe et al., 2011; Fell and Linn, 2013; Linn, 2008). Let $\delta_j \in \{0, 1\}$ denote the decision, with $\delta_j = 1$ if plant j switches, and let f_j denote the plant's fuel after the decision. Plants choose simultaneously.

Assumption 1 (Marginal Production Costs). Each plant j has a marginal cost of production that depends on its technology and on its fuel cost. Let $\ell \in \{\text{coal}, \text{coke}, \text{oil}, \text{gas}\}$ denote the fuel. The marginal cost under fuel ℓ is

$$c_{jt}^{\ell} = c\big(p_{jt}^{\ell}, X_j, q_{jt}\big), \tag{1}$$

where p_{jt}^{ℓ} is the price of fuel ℓ faced by plant j in period t, X_j collects plant technology, including capacity and kiln process, which determines the energy required per ton of clinker, and q_{jt} is output. Marginal cost rises with output as the plant approaches capacity.

Assumption 2 (Switching Costs). Switching to natural gas requires a one-time investment K_j, paid when the plant switches. It has a part that depends on observable plant characteristics and a part that the plant knows but I do not:

$$K_j \;=\; F + \beta_A A_j + \beta_d \log(1 + d_j) - \gamma N_j + \nu_j, \qquad \nu_j \sim N(0, \sigma^2). \tag{2}$$

F is the fixed cost of a conversion: burner and kiln modifications, a gas train, permits. $A_j \in \{0, 1\}$ indicates that the plant burned natural gas before the shale boom; such plants have burner compatibility, a supply contract and operating experience, so $\beta_A \leq 0$. d_j is the distance in miles from the plant to the nearest natural gas transmission pipeline, and $\beta_d \geq 0$ is the cost of the lateral that connects the plant to the network.[6] N_j is the present value of the air pollution compliance cost that switching to gas avoids, described next, and $\gamma \in [0, 1]$ is its weight. ν_j collects what the plant observes when it decides and I do not: the route of the lateral, permitting, site conditions and local opposition. It is independent of the observed shifters.

[5] Among all plants observed during the sample period, only one plant uses natural gas as its primary fuel throughout. I exclude this plant from the analysis sample.

[6] The distance term is a local approximation to the cost of pipeline access over the straight-line distance (Donaldson and Hornbeck, 2016); Scott (2023) uses a similar specification. The log form lets the cost of the first miles exceed that of later miles, and is supported by the reduced-form evidence that adoption declines with distance to the network.

NESHAP compliance. Plants also face the Portland Cement National Emission Standards for Hazardous Air Pollutants (NESHAP), which since 2010 limit mercury, hydrochloric acid, hydrocarbons and particulate emissions from kilns. Mercury is the limit that fuel choice affects: coal carries mercury and natural gas does not, so a plant that switches may avoid the activated carbon injection system the rule otherwise requires. EPA's engineering cost reports give the cost of that system as a function of capacity and kiln technology. N_j is the present value of this avoided cost. Because fuel switching was not evaluated as a compliance route in the regulatory reports, I do not impose $\gamma = 1$; γ measures how much of the engineering cost plants count when they decide, and Section 4.4 treats it as a parameter.

Present value of profits. Let $\mathbf{f} = (f_j, \mathbf{f}_{-j})$ denote the fuel profile of all plants, where $\mathbf{f}_{-j}$ is the profile of j's competitors. The second stage, described next, gives plant j's equilibrium shares $s_{jnt}(\mathbf{f})$ and prices $p_{jnt}(\mathbf{f})$ in each market n and its output $Q_{jt}(\mathbf{f})$. The present value of operating profit under fuel f_j over a horizon of T years at discount factor $\rho = (1+r)^{-1}$ is

$$\Pi_j(f_j, \mathbf{f}_{-j}) \;=\; \sum_{t=1}^{T} \rho^t \Big[\sum_{n \in \mathcal{N}} s_{jnt}(\mathbf{f})\, M_{nt}\, p_{jnt}(\mathbf{f}) \;-\; C_{jt}\big(Q_{jt}(\mathbf{f}); f_j\big) \Big], \tag{3}$$

where M_{nt} is the market size and $C_{jt}(\cdot\,; f_j)$ is the plant's variable cost under fuel f_j, the integral of the marginal cost in equation (1). The gain from switching is

$$V_j(\mathbf{f}_{-j}) \;=\; \Pi_j(\text{gas}, \mathbf{f}_{-j}) - \Pi_j(f_j^0, \mathbf{f}_{-j}), \tag{4}$$

where f_j^0 is the plant's current fuel. Switching changes the plant's fuel cost, and, under a green procurement standard, the set of buyers it may serve and the rivals it faces. The gain therefore depends on the fuels of the other plants. When plants in a multi-plant firm share buyers, the gain is summed over the firm's plants, because bids are set at the firm level.

The switching decision. Let $\mathcal{J}_j$ be plant j's information set when it decides. It knows its cost K_j, including ν_j, and forms an expectation of the gain. Plant j switches if

$$\mathbb{E}\big[V_j(\mathbf{f}_{-j}) \mid \mathcal{J}_j\big] \;\geq\; K_j. \tag{5}$$

A fuel profile in which equation (5) holds for every switcher and fails for every non-switcher, each taking the others' fuels as given, is a pure-strategy Nash equilibrium of the switching game. I use the condition in two ways. In estimation, it is the necessary condition that no plant gains from a one-step deviation from the observed profile (Pakes et al., 2015),

which identifies the switching cost without taking a stand on which equilibrium is played when several exist. In the counterfactuals, I solve for an equilibrium profile by iterated best response, letting each plant revise its decision given the others' until no plant wants to change (Lee and Pakes, 2009; Wollmann, 2018).

3.2 Second Stage: Compete in the Scoring Auction

Taking fuel choices as given, I first characterize a benchmark procurement environment, building on Miller et al. (2014b) and Miller et al. (2025). In this benchmark setting, plants compete for contracts under a scoring auction with a representative buyer. I then extend the environment to allow for two buyer types, green and conventional buyers, to study the effects of green procurement policy in Section 3.3

Net Utility and the Scoring Rule

Suppose buyers evaluate suppliers according to a score/net utility that depends on three components: (i) the transportation cost of procuring from that supplier, (ii) supplier-specific attributes such as import status, and (iii) the supplier's bid. The score function/net utility that buyer i in market n derives from supplier j submitting bid b_j is:

$$S_{ij} = -\beta\tau_{ij} + \boldsymbol{\gamma}'\mathbf{d}_j - \phi\, b_{ij} + \epsilon_{ij}, \tag{6}$$

where $\tau_{ij} \geq 0$ is the transportation cost between supplier j and buyer i; $\mathbf{d}_j$ is a vector of supplier-specific attributes (e.g., an indicator for import status); and $\phi > 0$ is the price parameter that converts monetary bids into utility units. ϵ_{ij} denotes the idiosyncratic buyer preference shocks. Equation (6) models a buyer's ranking of suppliers. A supplier with lower transportation costs, more favorable attributes, and a lower bid receives a higher score.

Assumption 3 (Information) The information structure is as follows:

(i) **Buyers:** Each buyer i has perfect information about the value of ϵ_{ij}.

(ii) **Suppliers:** Each supplier j has perfect information about the value of ϵ_{ij}.

(iii) **Common knowledge:** Each firm f has perfect information about the distribution, He, of the buyer/plant-level shocks for other firms.

Procurement Payoffs

Firms submit plant-level bids simultaneously. Buyer i in market n observes the vector of bids $\{b_{ij}\}_{j\in\mathcal{J}_n}$ from all plants. Buyer i evaluates each bidding plant j according to a scoring rule. The winning plant is the one that achieves the highest score among all eligible plants in the buyer's choice set. Conditional on winning, the transaction price is determined so that the buyer is indifferent between accepting the winner's offer and switching to the best available alternative. Formally, the following assumption introduces the second score auction.

Assumption 4 (Second-Score Auction)

(i) Buyer i awards the contract to the supplier with the highest score/net utility:

$$j^* \;=\; \arg\max_{j\in\mathcal{J}} \big\{ S_{jn}(b_j) \big\}. \tag{7}$$

(ii) The transaction price p_{ij^*} is set so that buyer i is indifferent between the winner at p_{ij^*} and the second-best alternative at its bid:

$$\mathcal{S}_{j^*n}(p_{ij^*}) = \max_{k\in\mathcal{J}_i\setminus\{j^*\}} \big\{ \mathcal{S}_{kn}(b_k) \big\}. \tag{8}$$

Solving equation (8) for the implied transaction price p_{ij^*}:

$$p_{ij^*} = \frac{1}{\phi}\Big[-\beta\tau_{j^*n} + \boldsymbol{\gamma}'\mathbf{d} + \epsilon_{ij*} - \max_{k\in\mathcal{J}_i\setminus\{j^*\}} \{ -\beta\tau_{jn} + \boldsymbol{\gamma}'\mathbf{d} - \phi b_{ik} + \epsilon_{ik} \} \Big]. \tag{9}$$

Equilibrium

Theorem 1 (Equilibrium Bidding)**.** *The following constitutes a Nash equilibrium:*

$$b_j \;=\; \begin{cases} c_j & \text{if } \bar{S}_{jn}(c_j) \geq 0, \\ \infty & \text{otherwise,} \end{cases} \tag{10}$$

for all $j \in \mathcal{J}$, $f \in \mathcal{F}$, and all buyers i. That is, each supplier bids at its marginal cost.

In equilibrium, the markup earned by the winning supplier j^* from buyer i is:

$$m_{ij^*} \;\equiv\; p_{ij^*} - c_{j^*} \;=\; \frac{1}{\phi}\Big[\mathcal{S}_{ij^*n} - \max_{k\neq j^*} \big\{ S_{ikn} \big\} \Big]. \tag{11}$$

The markup equals the net-utility advantage of the winning supplier over the best alternative,

scaled by $1/\phi$.

Note, substituting equation (6) into equation (11), the markup is:

$$m_{ij^*} \;=\; \frac{1}{\phi}\Big[\big(-\beta\tau_{j^*n} + \boldsymbol{\gamma}'\mathbf{d}_{j^*} - \phi c_{j*}\big) - \max_{k\neq j^*}\big(-\beta\tau_{kn} + \boldsymbol{\gamma}'\mathbf{d}_k - \phi c_k\big)\Big].$$

A supplier that is geographically proximate to a buyer relative to its rivals has a transportation cost advantage that translates directly into a higher markup ($\partial m_{ij^*}/\partial\tau_{j^*n} < 0$). Conversely, competitive pressure from a rival with a high equilibrium score, owing to either low cost or favorable location, compresses the winner's markup ($\partial m_{ij^*}/\partial c_k > 0$, $\partial m_{ij^*}/\partial\tau_{kn} > 0$).

3.3 Counterfactual Analysis: The Buy Clean Policy

This section introduces the Buy Clean policy into the procurement model. Two institutional features of the Buy Clean Initiative inform the modeling choices. First, the policy creates heterogeneous procurement environments, which I capture by introducing *green buyers* and *conventional buyers* in the procurement auction framework. Second, Buy Clean policies establish an embodied-carbon framework that defines the maximum allowable carbon intensity for targeted construction products. I model the embodied carbon threshold as an eligibility constraint that restricts the supplier choice set. This relates to the literature on bid preferences (Krasnokutskaya and Seim, 2011; Marion, 2007; Marion, 2017; Nakabayashi, 2013), where bidders choose the number or composition of subcontractors to meet procurement preference thresholds.

An alternative approach would be introducing embodied carbon into the procurement scoring function as a product attribute. However, this would require observing product-level heterogeneity and buyer-type-specific procurement decisions. Since the cement data are observed at the aggregate market level and do not contain product-level bids or buyer-type identifiers, the threshold formulation is both institutionally consistent and econometrically tractable. Formally:

Buyer types. In each market m, there are two types of buyers: *green buyers* and *conventional buyers*. Let $\lambda_m \in [0,1]$ denote the share of green buyers in market m. Green buyers procure under Buy Clean rules, while conventional buyers procure from the unrestricted supplier set. The composition λ_m may vary across markets, reflecting heterogeneity in local procurement policy, public construction activity, and private low-carbon purchasing commitments.

Embodied-carbon intensity. Each plant j produces cement with embodied-carbon intensity $e_j(f_j) \geq 0$. Embodied-carbon intensity depends on the plant's fuel choice $f_j \in \{\text{coal}, \text{gas}\}$. Switching from coal or petroleum coke to natural gas reduces embodied-carbon intensity because natural gas has lower combustion emissions per unit of energy. I write $e_j^{\text{coal}} > e_j^{\text{gas}} \geq 0$ to denote embodied-carbon intensity under coal and gas, respectively.

Buy Clean threshold and eligible supplier set. Each green buyer in market m applies an embodied-carbon threshold $\bar{e}_m > 0$. A plant is eligible for green procurement in market m if its embodied-carbon intensity does not exceed the threshold. Given a fuel profile $\mathbf{f} = \{f_j\}_{j \in \mathcal{J}_m}$, the eligible supplier set is

$$\mathcal{G}_m(\mathbf{f}; \bar{e}_m) = \{j \in \mathcal{J}_m : e_j(f_j) \leq \bar{e}_m\},$$

where $\mathcal{J}_m$ is the full set of suppliers that can serve market m. A lower $\bar{e}_m$ represents a more stringent Buy Clean standard.

The counterfactual analysis holds the Buy Clean threshold as given and studies how the resulting restriction of the supplier set affects procurement outcomes. The following two assumptions ensure that the green procurement auction remains well-defined.

Assumption 5 (Minimum Competition) In every local market m, the Buy Clean threshold is such that at least two suppliers are eligible.

Assumption 5 rules out the case in which the Buy Clean threshold leaves only one eligible supplier in a market. This restriction is consistent with the policy environment because procurement agencies would allow waivers when a standard would leave only one feasible supplier.[7] With at least two eligible suppliers, the equilibrium submitted bids remain equal to marginal cost in the second-score auction.

Competition Effects of the Buy Clean Eligibility Rule

The Buy Clean policy affects procurement competition by changing the set of suppliers that can compete for green buyers. Conventional buyers continue to face the full supplier set J_m, while green buyers face only the eligible supplier set $G_m \subseteq J_m$. Holding fuel choices and

[7] For example, the Maryland Buy Clean Act allows the Department of General Services to waive certain requirements if compliance would be technically infeasible, result in a significant increase in project cost, result in a significant delay in project completion, or result in only one source or manufacturer being able to provide the necessary materials.

marginal costs fixed, this section characterizes how the restriction of the supplier set affects transaction prices, markups, and market shares.

In the second-score auction, equilibrium submitted bids remain equal to marginal cost. Thus, Buy Clean does not directly raise submitted bids. Instead, the transaction price changes because the best alternative available to the buyer changes. Recall that under the scoring rule, the transaction price p_{ij^*m} is set so that buyer i is indifferent between the winning supplier j^* at price p_{ij^*m} and the second-best alternative at its equilibrium bid. If the buyer faces supplier set $\mathcal{J}$, the transaction price is

$$p_{ij^*m}(\mathcal{J}) = \frac{1}{\phi}\left[-\beta\tau_{j^*n} + \boldsymbol{\gamma}'\mathbf{d}_{j^*} - \max_{k\in\mathcal{J}\setminus\{j^*\}}\left\{-\beta\tau_{kn} + \boldsymbol{\gamma}'\mathbf{d}_k - \phi c_k\right\}\right]. \tag{12}$$

For conventional buyers, $\mathcal{J} = J_m$. For green buyers, $\mathcal{J} = G_m$. Because $G_m \subseteq J_m$, the best alternative facing a green buyer is weakly worse:

$$\max_{k\in G_m\setminus\{j^*\}}\left\{-\beta\tau_{kn} + \boldsymbol{\gamma}'\mathbf{d}_k - \phi c_k\right\} \leq \max_{k\in J_m\setminus\{j^*\}}\left\{-\beta\tau_{kn} + \boldsymbol{\gamma}'\mathbf{d}_k - \phi c_k\right\}.$$

Therefore, for an eligible supplier j^* that wins under both supplier sets,

$$p_{ij^*m}(G_m) \geq p_{ij^*m}(J_m).$$

The inequality is strict when the competitor excluded by the Buy Clean threshold would have been the strongest alternative in the conventional auction. In this case, the green buyer has a weaker outside option, so the eligible supplier receives a higher transaction price.

The markup is the transaction price net of marginal cost $\mu_{ij^*m}(\mathcal{J}) = p_{ij^*m}(\mathcal{J}) - c_{j^*}$. Thus, Buy Clean therefore raises markups when the eligibility rule excludes a close competitor:

$$m_{ij^*m}(G_m) \geq m_{ij^*m}(J_m).$$

The policy also reallocates market shares across buyer segments and local markets. Ineligible suppliers cannot serve green buyers. Eligible plants may sell more to green buyers while selling less to conventional buyers or to other local markets. Therefore, Buy Clean changes both the composition of buyers served and, depending on capacity constraints and import competition, the total quantity sold by each plant.

Thus, Buy Clean affects profits through both the markup and market-share margins. Eligible suppliers may serve a larger share of the green procurement segment, and conditional

on winning, they may receive higher transaction prices and higher markups because the strongest excluded competitors are no longer eligible for the green market.

4 Empirical Specification

4.1 Second Stage: Auction

I follow Miller et al. (2025) and assume a nested-logit demand system to model cement procurement. Let m index county markets, j index suppliers, and t index years. Several assumptions are needed for estimation under regional aggregated data with county-level moment conditions. The first assumption is that the observable gross utility u_{mjt} is the same within a county market-year. The second assumption is that all cement suppliers belong to one inside nest, while the outside goods belong to a separate outside nest at a constant marginal cost.

Cement suppliers include both domestic cement plants and import terminals. Buyer i in county m receives gross utility from supplier j in year t:

$$u_{ijt} = u_{mjt} + \zeta_{it} + (1-\sigma)\varepsilon_{ijt}, \tag{13}$$

where u_{mjt} is the component of gross utility common to buyers in county m, ζ_{it} is the buyer-specific component common to all cement suppliers in year t. ε_{ijt} is i.i.d. Type-I Extreme Value, and $\sigma \in [0, 1)$ is the nesting parameter. The outside good provides utility $(1-\sigma)\tilde{\varepsilon}_{i0t}$, with its mean utility normalized to zero. I set $\sigma = 0.9$, following Miller et al. (2025), and report robustness checks using alternative values of σ.

The common component of gross utility is specified as:

$$u_{m(i)jt} = \alpha_0 + \alpha_1 \tilde{t} - \alpha_2 b_{jt} - \alpha_3 \tau_{jm} + \gamma_1 \text{IMPORT}_j + \gamma_2 \text{IMPORT}_j \times \tilde{t}, \tag{14}$$

where τ_{jm} denotes the shipment cost from supplier j to county m, and α_3 captures the disutility of shipment costs. b_{jt} denotes the bid value from supplier j. IMPORT_j is an indicator equal to one if supplier j is an import terminal. $\tilde{t} = t - \bar{t}$ is a demeaned time trend.

Bid

For domestic plants, the bid equals the plant-level marginal cost, which is derived from a kiln-level cost function that accounts for fuel costs and capacity constraints. For import

terminals, the bid equals the observed customs value of imported cement at the terminal, inclusive of insurance, freight, and delivery charges to the port of entry.

Specifically, following cement literature capturing cost structure (Ryan, 2012; Fowlie et al., 2016; Miller et al., 2025), assume the marginal cost of production at any kiln ℓ operated by plant j in year t is:

$$c_{jt}^{(\ell)}\left(Q_{jt}^{(\ell)}\right) \;=\; \beta_0 + \beta_1 \,\text{FuelCost}_{jt}^{(\ell)} + \beta_2\,\tilde{t} \;+\; \beta_3\left(\frac{Q_{jt}^{(\ell)}}{\text{CAP}_{jt}^{(\ell)}} - \nu\right)^2 \mathbf{1}\left\{\frac{Q_{jt}^{(\ell)}}{\text{CAP}_{jt}^{(\ell)}} > \nu\right\}, \tag{15}$$

where $Q_{jt}^{(\ell)}$ is kiln-level output, $\text{CAP}_{jt}^{(\ell)}$ is annual kiln capacity, and $\text{FuelCost}_{jt}^{(\ell)}$ is the fuel cost per tonne for the kiln ℓ, constructed from state-level fossil fuel prices and kiln-level energy requirements. The Appendix A3.1 provides details for computing fuel costs. The parameters $\beta_0 > 0$, $\beta_1 > 0$, and β_2 govern, respectively, a constant component of marginal cost (labor and maintenance), the pass-through of fuel costs, and a common time trend. Kiln-level marginal cost is flat at low utilization rates and rises quadratically once the utilization rate $Q_{jt}^{(\ell)}/\text{CAP}_{jt}^{(\ell)}$ exceeds the threshold $\nu \in (0,1)$, with the penalty scaled by $\beta_3 > 0$.

We assume plants allocate output across kilns to minimize total variable cost. For a given total plant output Q_{jt}, the optimum equates the marginal cost across all operated kilns, yielding a continuous, weakly increasing plant-level marginal cost function $c_{jt}(Q_{jt})$.

To summarize, the bid is:

$$b_{jt} \;=\; \begin{cases} c_{jt}(Q_{jt}^*) & \text{if } j \text{ is a domestic plant,} \\ \text{Customs Value}_{jt} & \text{if } j \text{ is an import terminal,} \end{cases} \tag{16}$$

where Q_{jt}^* is the equilibrium quantity sold by plant j in year t, obtained as part of the fixed-point solution described in Section 4.3. Because $c_{jt}(\cdot)$ is non-decreasing, domestic bids rise with output, linking the cost side and the demand side through the fixed-point equilibrium.

Transportation cost

Transportation costs $\tau_{jm} \geq 0$ capture the cost of shipping one unit of cement from supplier j to buyer county m, measured in dollars per tonne. I model two transportation modes:

land-based transport (truck or rail) and water-based transport (barge and coastal shipping).

$$\tau_{jm} = \begin{cases} \gamma_L d_{jm}, & \text{if shipment is overland,} \\ \gamma_0 + \gamma_L \left(d_{j,p(j)} + d_{p(m),m} \right) + \gamma_W d^{\text{water}}_{p(j),p(m)}, & \text{if shipment is waterborne.} \end{cases} \tag{17}$$

where I assume the overland transportation cost is linear in the distance of the routed overland shipment d_{jm}. $\gamma_L > 0$ is the marginal cost per mile of overland transport. Waterborne transportation involves three components: overland distance $d_{j,p(j)}$ from the plant to its nearest port $p(j)$, navigation along the waterway $d^{\text{water}}_{p(j),p(m)}$ from $p(j)$ to the destination port $p(m)$, and overland delivery $d_{p(m),m}$ from $p(n)$ to the buyer market. $\gamma_0 \geq 0$ is a fixed cost of accessing the waterway system, and $\gamma_W < \gamma_L$ reflects the lower marginal cost per mile of waterborne transport. Transportation costs are assumed to be time-invariant. For each plant-market pair, the realized transportation cost is the minimum across the two modes.

I calibrate the cost parameters $(\gamma_0, \gamma_L, \gamma_W)$ using data from the Commodity Flow Survey (CFS). Identification comes from cross-sectional variation in shipment distance, port access, and waterway distance across plant-market pairs. I assume the route geography affects mode choice through transportation costs and is uncorrelated with unobserved shipment factors, such as delivery urgency, port congestion, or local carrier availability. Because port locations and waterway networks are largely shaped by physical geography and historical infrastructure, I treat this variation as predetermined with respect to unobserved shocks.

Specifically, I group observed shipments into bins by total shipment distance, and within each bin, I compute the empirical share of shipments for both modes. I then recover the parameter vector via grid search, choosing values that minimize the sum of squared distances between the model-predicted mode share and the empirical share across distance bins. This approach ensures that the calibrated cost parameters reproduce the broad spatial pattern of mode choice observed in the data without requiring an explicit structural model of shipper optimization. I assume transportation costs do not vary across years. Appendix **??** presents full details of the calibration procedure, including port assignment, waterway network construction, and grid search implementation.

Import indicator

IMPORT_j equals one if supplier j is an import terminal and zero if it is a domestic plant. The parameter γ_1 captures the average net utility difference between imported and domestic cement, conditional on transportation costs and bids. This term allows buyers to differently

value imported cement from domestic cement due to unobserved attributes such as perceived quality differences, project specifications, reliability of supply, or preferences related to diversification of supply sources. The interaction $\text{IMPORT}_j \times \tilde{t}$ allows the relative competitiveness of imported cement to evolve over time. Appendix A3.2 presents details of constructing the set of import districts.

Market Size

I construct the county-year market size as a proxy for downstream cement demand. Using the BEA Input–Output Total Requirements Table, I identify cement-consuming sectors, mainly construction and mining, and combine BEA sector-level value added with QCEW county-level employment shares to allocate demand across counties and years. I then use USGS regional cement consumption data to scale the county-level measure so that the aggregate predicted market size matches observed regional cement consumption. Appendix A3.3 presents the details of the calculation.

4.2 Equilibrium Outcomes

Let $\bar{u}^*_{jmt}$ denote the mean net utility. I there compute the following closed-form expressions for market, average price, and markup over the idiosyncratic preference shock.

Market shares.

$$s^*_{jmt} \;=\; \frac{\exp\left(\dfrac{\bar{u}^*_{jmt}}{1-\sigma}\right)}{\displaystyle\sum_{k\in\mathcal{J}_t}\exp\left(\dfrac{\bar{u}^*_{kmt}}{1-\sigma}\right)} \;\cdot\; \frac{\mathcal{IV}^{1-\sigma}_{mt}}{1+\mathcal{IV}^{1-\sigma}_{mt}}, \qquad \mathcal{IV}_{mt} \;\equiv\; \sum_{k\in\mathcal{J}_t}\exp\left(\frac{\bar{u}^*_{kmt}}{1-\sigma}\right) \tag{18}$$

Outside good share.

$$s^*_{0mt} \;=\; \frac{1}{1+\mathcal{IV}^{1-\sigma}_{mt}} \tag{19}$$

Markup. Each plant j is owned by company c. Let $s^*_{cmt} \equiv \sum_{k\in\mathcal{J}^c_t} s^*_{kmt}$ denote the total market share of all plants owned by firm c in county m in period t. Then:

$$m^*_{jmt} \;=\; -\frac{1}{\phi}\cdot\frac{1}{F^*_{cmt}}\cdot\log\left[1-(1-s^*_{0mt})\left(1-\left(1-\frac{s^*_{cmt}}{1-s^*_{0mt}}\right)^{1-\sigma}\right)\right] \tag{20}$$

Price and buyer surplus.

$$p^*_{jmt} \;=\; m^*_{jmt} + b_{jt} \tag{21}$$

Note that transaction prices vary across plants of the same firm because fuel costs b_{jt} differ across plants, even though markups m^*_{jmnt} are identical for co-owned plants in the same county.

4.3 Estimation

We estimate the parameter vector $\theta = (\alpha_0, \alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2, \beta_0, \beta_1, \beta_2, \beta_3, \nu)$ by nonlinear least squares (NLS), minimizing the squared distance between model predictions and observed aggregate outcomes (Miller and Osborne 2014b; Elickson et al., 2020). The objective function is:

$$\mathcal{L}(\theta) \;=\; \kappa_{\text{price}} \cdot \mathcal{L}^{\text{price}}_t(\theta) \;+\; \kappa_{\text{prod}} \cdot \mathcal{L}^{\text{prod}}_t(\theta) \tag{22}$$

The moment conditions are chosen to match the two primary aggregate outcomes observed in the data, regional cement prices and regional production quantities, in order to recover the model parameters. The moment conditions are:

$$\mathcal{L}^{\text{price}}_t(\theta) \;=\; \frac{1}{R_t} \sum_{r=1}^{R_t} \big(p^{\text{data}}_{rt} - p^{\text{pred}}_{rt}(\theta)\big)^2, \tag{23}$$

$$\mathcal{L}^{\text{prod}}_t(\theta) \;=\; \frac{1}{R^{\text{prod}}_t} \sum_{r=1}^{R^{\text{prod}}_t} \big(Q^{\text{data}}_{rt} - Q^{\text{pred}}_{rt}(\theta)\big)^2, \tag{24}$$

where $\kappa_m = 1/\text{Var}(y_m)$ is the inverse variance of moment m pooled across all years, ensuring each moment type contributes comparably to the total loss regardless of scale.

Predicted regional production $Q^{\text{pred}}_{rt}(\theta)$ is the total quantity sold by all plants located in production region r, aggregated across all buyer counties:

$$Q^{\text{pred}}_{rt}(\theta) \;=\; \sum_{j \in r} \sum_{n \in \mathcal{N}} s^*_{jnt}(\theta) \cdot M_{nt}. \tag{25}$$

Predicted regional prices $p^{\text{pred}}_{rt}(\theta)$ are the quantity-weighted average predicted prices across

all plant–county pairs (j, n) associated with price region r:

$$p_{rt}^{\text{pred}} = \frac{\sum_{j \in r} \sum_{n \in r} Q_{jnt}^{*} \, p_{mjt}^{*}}{\sum_{j \in r} \sum_{n \in r} Q_{mjt}^{*}}, \tag{26}$$

where

$$Q_{jnt}^{*} = s_{jnt}^{*} \cdot M_{nt}.$$

The second-stage parameters are estimated by matching observed regional production quantities and prices to county-level market shares and markups, using the spatial concentrated structure of cement markets. The main identifying variation comes from the following sources. First, pre-existing transportation geography creates cross-sectional variation in delivered costs across plant-county pairs. A potential concern is endogenous plant entry and exit: plants may enter near high-demand markets, making distance correlated with unobserved demand shocks. This concern is limited because the plant network is mostly fixed. Among 114 plants during the sample period, only three plants entered, and less than 15% exited, with most exits occurring around 2008, likely due to the financial crisis demand shock. I exclude entering and exiting plants as a robustness check.

Second, I use state-level variation in fuel prices and cross-plant variation in kiln technology, capacity, and predetermined fuel types to shift cost-based bids across plants and years. The identifying assumption is that these cost shifters affect procurement outcomes through marginal costs, not through unobserved demand shocks. The main concern is that technology and capacity may be endogenous if plants expand or improve energy efficiency in response to local demand growth. This concern is limited to kiln technology because most energy-efficiency upgrades occurred before the sample period, and fewer than 5% of plants undertake such upgrades during 2002-2019. Capacity expansion is more relevant: about 23% of plants expand capacity during the sample period. I therefore address potential endogenous capacity adjustment by showing robustness to excluding plants with capacity expansions.

Third, variation in exposure to import terminals helps identify the relative competitiveness of imported cement, conditional on bids and transportation costs. I restrict the import choice set to the 20 import terminals that account for about 90% of U.S. cement imports, so that the model focuses on economically relevant import suppliers. The main concern is that import customs values may be endogenous to local demand shocks. As a robustness check, I replace current import customs values with lagged customs values.

Fourth, variation in county-year market size helps pin down total cement demand and the outside option. Market size is constructed from the cement requirement index and value added in the broad downstream sectors, mainly construction and mining. Since cement is the only input into these sectors, cement suppliers are unlikely to affect this market-size measure in reverse. This reduces the concern that market size is mechanically determined by cement supply shocks.

4.4 First Stage: Fuel-Switching Investment

This section estimates the fuel-switching investment model. The estimation covers the period from 2006 to 2019. I exclude observations from 2020 onwards because: (1) the COVID-19 pandemic created significant disruptions to cement production and investment decisions; and (2) the post-2020 period marks a distinct policy regime characterized by federal Buy Clean initiatives, Inflation Reduction Act subsidies, and industry-wide net zero commitments, which altered the incentive structure for fuel switching. The 2006-2019 window represents a coherent regime in which switching decisions were primarily driven by the shale gas revolution and compliance with air quality regulation (USGS, 2015).

Rather than imposing a parametric distribution on the unobserved component of switching costs, I follow the moment inequality approach of Dickstein and Morales (2018) along with the revealed preference conditions of Pakes et al. (2015). The literature in moment inequalities differs on how it handles the sunk-cost component that firms observe, but the econometrician does not. Wollmann (2018), for example, assumes this error component is common to all firms within a product type in a given year. Selection then determines how many firms act rather than which firms act, and weighting the observations so that each year's shock enters once allows the shock to average out over a long panel of product entry and exit decisions.

In the cement industry, the cost of converting a kiln to natural gas depends on site-specific factors, such as the route of a pipeline lateral, permitting, and local opposition, that a plant knows when it decides but an econometrician does not observe, so the plants that switch are selected on low realizations of this cost. Each plant makes a single, largely irreversible decision within a few-year window, and no long time series can average out a common shock.

Dickstein and Morales (2018) address this setting: a binary sunk investment whose cost includes a firm-specific component known to the firm, and expected returns formed from an information set the econometrician does not fully observe. Their odds-based and revealed preference inequalities integrate over a parametric distribution of the plant-specific cost shock and remain valid for any plant information set that contains the instruments. As a

result, discrepancies between my measure of the expected gain from switching and plants' own expectations, including their expectations of post-shale fuel prices, do not invalidate the bounds. The cost of this approach is a distributional assumption on the plant-specific shock, which I take to be normal and independent of the observed cost shifters.

Using the estimated second-stage procurement model, I compute the present value of the operating profit difference between the natural gas and baseline fuel scenarios:

$$V_j = \sum_{\tau=t}^{T_j} \rho^{\tau-t} \sum_{n \in \mathcal{N}} \left[\hat{s}_{jn\tau}^{gas} \hat{m}_{jn\tau}^{gas} - \hat{s}_{jn\tau}^{base} \hat{m}_{jn\tau}^{base} \right] M_{n\tau}, \tag{27}$$

where $\hat{s}_{jn\tau}$ and $\hat{m}_{jn\tau}$ are the predicted market share and markup from the second-stage model, $M_{n\tau}$ is county-level market size, and T_j is the expected remaining operating horizon at the time of switching. Both components are evaluated at the plant's decision year and discounted over the kiln's remaining lifespan $T_j = 60 - \text{kiln_age}_j$[8] at discount rate $r = 0.05$. I consider alternative options for the payback year, including 5 years and 10 years. I assume plants switch fully to pure natural gas. Another switching scenario for robustness checks assumes plants partially adopt natural gas while retaining existing fuel sources as a multi-fuel portfolio.

The net switching cost is parameterized as:

$$K_j \;=\; \beta' x_j + \nu_j = F + \beta_A A_j + \beta_d \log(1 + l_j) + \gamma\, N_j + \nu_j, \qquad \nu_j \sim N(0, \sigma^2), \tag{28}$$

where F is the fixed switching cost, l_j is the plant's distance to the nearest natural gas transmission pipeline, A_j indicates pre-existing pipeline access before 2006, and N is the annual NESHAP engineering-based control cost that switching to gas avoids, a function of the plant's capacity and kiln technology, based on EPA policy documents. The parameter γ captures the extent to which NESHAP compliance pressure increases the incentive to switch; because fuel switching was not explicitly evaluated as a NESHAP compliance option in the regulatory cost reports, γN should be interpreted as a reduced-form measure of NESHAP-related switching incentives rather than a directly observed avoided compliance cost. Two errors separate what the plant knows from what I observe. The first is the plant's forecast error: my measure V_j is a realized gain from the estimated model, and it differs from the gain the plant expected when it decided. The second is ν_j, the part of the switching cost the plant knows and I do not. Pakes et al. (2015) allow the first and not the second; a

[8]PIS data shows that the average retirement age is 48. I take the maximum kiln age, 60, shown in the data as a measure of kiln lifespan. I consider different lifespan years in the robustness check.

probit allows the second and not the first. Dickstein and Morales (2018) allow both, under three assumptions. First, the forecast error has mean zero given the plant's information: $\mathbb{E}[V_j - \mathbb{E}(V_j \mid \mathcal{J}_j) \mid \mathcal{J}_j] = 0$. Second, I observe a vector $z_j \subseteq \mathcal{J}_j$ of variables the plant knew when it decided. Third, ν_j is known to the plant, normally distributed with mean zero and standard deviation σ, and independent of z_j and x_j. Normality is not essential: any log-concave distribution gives the same inequalities (Kline, Pakes, and Tamer, 2021).

σ is the dispersion of the cost component I do not observe. At $\sigma = 0$, the observables x_j explain the cost fully, and the only error is the plant's forecast error. A large σ means that private information about the cost dominates, so that the gain alone predicts switching poorly, and that the observed switchers are plants that drew a low cost.

Moment inequalities. Let $d_{jt} = 1$ if the plant switched and $\pi_{jt} = V_{jt} - \beta' x_{jt}$ be the realized gain net of the observable cost. Let ϕ and Φ be the standard normal density and distribution, and $\lambda(a) = \phi(a)/\Phi(a)$. For any nonnegative function $h(z_{jt})$, the assumptions above imply two pairs of inequalities Dickstein and Morales, (2018):

$$\text{(RP)} \quad \mathbb{E}\big[h(z)\{d\,\pi + (1-d)\,\sigma\,\lambda(-\pi/\sigma)\}\big] \;\geq\; 0, \qquad \mathbb{E}\big[h(z)\{-(1-d)\,\pi + d\,\sigma\,\lambda(\pi/\sigma)\}\big] \;\geq\; 0, \tag{29}$$

$$\text{(OB)} \quad \mathbb{E}\Big[h(z)\Big\{d\,\frac{1-\Phi(\pi/\sigma)}{\Phi(\pi/\sigma)} - (1-d)\Big\}\Big] \;\geq\; 0, \qquad \mathbb{E}\Big[h(z)\Big\{(1-d)\,\frac{\Phi(\pi/\sigma)}{1-\Phi(\pi/\sigma)} - d\Big\}\Big] \;\geq\; 0. \tag{30}$$

The revealed preference pair (RP) describes that a switcher's gain must cover its cost, and a stayer's must not. But a switcher is a plant that drew a low-cost shock. Its ν averages $-\sigma\lambda(\pi/\sigma)$, not zero. The second term in each moment adds this selection back. Without it, the switchers' moment would understate the cost. The odds-based pair (OB) uses a second implication of the model: a plant switches with probability $\Phi(\pi/\sigma)$ when π is the plant's expectation. Each moment compares the observed odds of switching with the odds the model implies. The RP moments bound the level of the cost. The OB moments involve σ directly and are what bound it. At $\sigma = 0$, the RP pair reduces to the inequalities of Pakes et al. (2015), which I report alongside. Each relation holds with equality at the plant's expectation. I observe the realized gain, which contains the forecast error. The correction terms and the odds ratios are convex in π, so by Jensen's inequality the equality at the expectation becomes an inequality at the realized value. This is why the model yields bounds rather than point estimates.

$h(z)$ takes the values of indicators of variables known at the decision: prior gas use, terciles of the distance to the pipeline network, terciles of the gain evaluated at the decision year's prices. Each instrument turns an inequality into a statement about one group of plants. For example, the plants without prior gas and with a large gain that did not switch must have faced a cost above that gain.

The upper bound on the cost comes from the switchers: a plant that switched with gain V reveals that its cost was at most V, plus the selection term. The lower bound comes from the stayers with large gains: a plant that did not switch reveals that its cost was at least V, minus the selection term. σ is identified from the fit of the two bounds. At $\sigma = 0$ every switcher needs $V \geq K$ and every stayer $V \leq K$; switchers with negative gains and stayers with large gains then leave no K that fits, or only a wide range. As σ grows, the selection terms widen the room for each bound, but the odds-based moments require the implied switching probabilities to match the switch rates across the instrument cells. A σ that is too large predicts too many switches among high-gain stayers and too few among low-gain switchers. The data therefore bound σ from both sides.

I compute the 95 percent confidence set with the generalized moment selection test of Andrews and Soares (2010), as Dickstein and Morales (2018) do. At each point of a grid over (β, σ), the statistic is the sum of squared standardized negative sample moments. The critical value comes from a bootstrap that resamples plants and drops moments far from binding. The confidence set is the collection of grid points not rejected. The interval reported for a single coefficient, or for the cost of a group of plants, is the projection of the set: the smallest and largest value of that quantity over the accepted points. I report the constant-cost model and each shifter added to it, and the two-shifter model in the appendix.

5 Estimation Results

5.1 Procurement Competition

Table 1 reports preliminary parameter estimates. From the transportation cost calibration, overland transportation cost is \$0.15 per tonne-mile, while barge shipments involve a fixed loading cost of \$5 per tonne plus an additional \$0.05 per tonne-mile for water transport. This specification implies that overland shipments are preferred at short distances, while barge becomes relatively more attractive for longer hauls when waterways are available. The demand-side estimates also provide evidence of substantial geographic differentiation in cement procurement. The positive and significant transportation cost coefficient indicates that

Table 1: Second-Stage Estimates: Cement Procurement

Parameter	Coefficient	Estimate	S.E.
Gross Utility Parameters			
Constant	α_0	3.2541***	(0.2015)
Time Trend	α_1	0.1883***	(0.0299)
Bid	α_2	-0.0078***	(0.0008)
Transportation Cost	α_3	-0.2552***	(0.0474)
Import	γ_1	-0.3971***	(0.0544)
Import × Time Trend	γ_2	-0.0236***	(0.0040)
Marginal Cost Parameters			
Constant	β_0	26.4443***	(1.7300)
Fuel Cost	β_1	1.6182***	(0.1214)
Capacity Penalty	β_3	59.5379***	(1.7474)
Utilization Threshold	ν	0.3541***	(0.0169)
Pre-set Parameter			
Nesting Parameter	σ	0.9000	–

Notes: Standard errors are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The nesting parameter σ is fixed at 0.9. Estimation assumes single-plant firms.

delivered costs remain an important determinant of supplier choice, implying that procurement competition is fundamentally local. Buyers also exhibit a significant negative preference for imported cement relative to domestic suppliers, conditional on price and transportation costs, a preference that has grown more pronounced over time.

On the cost side, the estimates reveal substantial heterogeneity in production costs, driven by both energy expenditures and capacity utilization. The positive time trend in marginal costs indicates that production costs have risen over time, even after controlling for fuel prices. The fuel cost pass-through coefficient of 1.62 implies that a one-dollar increase in fuel cost raises marginal cost by \$1.62 per metric tonne, consistent with existing evidence on cost pass-through in the cement industry (Miller et al., 2017). The constant implies that other inputs, such as materials and labor, contribute \$26.44 to the marginal cost per metric tonne in the average year. The estimated utilization threshold and capacity penalty imply that marginal costs are highly convex in output. Marginal costs begin rising steeply once utilization exceeds 35.4% of capacity, implying that plants face binding capacity constraints at moderate output levels. Capacity Penalty indicates that producing at capacity increases the marginal cost by \$59.34 per metric tonne compared to producing at a utilization rate below 35%.

5.2 Fuel Switching

Table 6 reports the switching cost under each estimation approach, in the format of **dickstein2018**. The first three columns are probits that assume the plant's expectation is known: the realized gain, the gain at the decision year's prices, or the projection of the gain on the instruments. The fourth column uses the revealed preference inequalities alone, with no cost shock. The last column is the estimator of Section 4.4. Table 8 then reports the cost by group of plants.

The common cost is well identified. The confidence set is \$67 to \$82 million for a 10-year present value, or \$9 to \$11 million a year. For a plant of average size, 1.1 million tons, that is \$8 to \$10 per ton of clinker, against a fuel cost gap between coal and gas of \$5 to \$30 per ton over the period.

The five columns are ordered as Dickstein and Morales (2018) find. The revealed preference set in column (4) is the lowest, \$7 to \$50 million. It treats every switcher's gain as a bound on its cost, but a switcher is a plant that drew a low cost shock, so the bound is too low. The probits in columns (1) to (3) are the highest, \$147 to \$197 million. Each treats its measure of the gain as the plant's expectation, so every gap between that measure and the decision is read as cost dispersion. That inflates σ to \$94 to \$120 million, and a larger σ needs a larger K to match the low switch rate. The Dickstein–Morales set lies between. It corrects the selection in column (4) and does not force the plant's expectation to equal my measure.

Table 7 reports the set at each σ. The set is empty beyond $\sigma = 30$, and K rises with σ: the larger the private component of the cost, the more the observed switchers are plants that drew a low one, and the higher the average cost consistent with their switching. The data alone accept σ from 0 to 30. The 2022 holdout picks the range. At $\sigma = 0$ the sets predict 13 to 40 switches against 5 observed, because many plants had a gain above \$50 million and did not switch. At $\sigma = 15$ to 20 they predict 5.6 to 7.6. I take $\sigma = 15$ to 20 as the preferred range and report the rest alongside.

Table 8 reports the cost by group of plants, one shifter at a time. With 21 switches each shifter is bounded on one side. Prior gas use lowers the cost or leaves it unchanged: the cost ratio is at most 1.35, and the points of the set with the smallest test statistic put it near 0.4. Distance raises the cost or leaves it unchanged, with the smallest-statistic points at 1.3 to 1.7 per doubling. The period indicator is not bounded. Under definition 2, with 37 switches, the prior gas ratio is bounded on both sides, 0.18 to 0.74, and the distance ratio is at least 1.07 per doubling. The signs are those the engineering story predicts. The sets say the data cannot rule out that the effects are small.

The cost the decisions reveal is several times the engineering cost of the pipe. The lateral for the median plant costs $5 to $8 million at EPA's cost factor. Keystone Cement's Bath plant, one of the 2022 switchers, connected to a local distribution line for $2.2 million. The gap is the rest of what a plant gives up when it commits to gas: burner and kiln modifications, exposure to gas price risk, a hurdle rate above the 5 percent used to discount the gain, and the option value of waiting. The same ordering, estimated fixed costs well above accounting costs, is what Dickstein and Morales (2018) and Wollmann (2018) find.

5.3 Model Fit

Before moving to the policy counterfactual analysis, I assess my model's performance by examining its model fit to both targeted and untargeted moments.

I evaluate how closely the second-stage model matches the moments. Figure 9 in the Appendix, which compares observed and predicted procurement quantities aggregated at the production-region level. The model closely reproduces the observed distribution of regional sales, with most observations lying near the 45-degree line. While some large markets are underpredicted, the overall fit suggests that the model captures the primary determinants of regional procurement patterns and market allocation. Figure 10 in the Appendix compares observed and predicted regional cement prices. The model reproduces the broad cross-sectional and temporal variation in prices, with predicted prices closely increasing with observed prices across regions and years. While the model tends to smooth some of the extremely high- and low-price observations, the overall fit suggests that the estimated demand and cost parameters capture the primary determinants of regional pricing patterns.

6 Counterfactual Simulation

In this section, I use the estimated model to conduct counterfactual simulations evaluating the effects of alternative environmental policies. To do so, I first need to establish a link between plant fuel choice and cement carbon emissions. Specifically, I construct plant-level measures of per-ton cement CO_2 emissions using Global Warming Potential (GWP) data and relate these emissions to observed fuel mixes.

6.1 Global Warming Potential (GWP)

To link plant-level per-ton cement CO_2 emissions to fuel choice, I collect Environmental Product Declarations (EPDs) for cement products produced by U.S. cement plants. EPDs report

product-level Global Warming Potential (GWP), measured in kilograms of CO_2 equivalent per ton of cement, and provide a standardized measure of lifecycle emissions. The EPDs also report product type and clinker-to-cement ratio. I match these EPDs to plant characteristics and fuel-use information from the PIS and EIA datasets. I then construct an engineering-computed GWP for each plant based on its fuel mix and kiln technology. The engineering construction combines two emission sources scaled to a per-ton-cement basis. Combustion emissions are computed as the kiln energy intensity (MMBtu per ton clinker, varying by process technology: dry-with-precalciner, dry-with-preheater, plain dry, or wet) multiplied by the plant's fuel-mix-weighted emission factor, constructed from EPA standard factors per MMBtu for coal, petroleum coke, natural gas, oil, and alternative fuels. Process emissions are set at 540 kg CO_2 per ton clinker, the chemical yield from calcination of limestone. Both components are then scaled by the plant's clinker-to-cement ratio.

Figure 11 compares the engineering-computed GWP with the observed GWP reported in plant EPD documents. The two measures are positively correlated with a slope of 0.73, indicating that the engineering construction captures meaningful variation in plant-level cement emissions. Observed GWP values are systematically higher than the engineering estimates, with most observations lying above the 45-degree line and an estimated intercept of approximately 363 kg $CO2_2$ per ton cement. This level shift reflects emission sources outside the engineering-based clinker emission scope, including electricity consumption, upstream raw material and fuel transport, and finish grinding energy.

Because EPD coverage is incomplete, I impute GWP for plants without published EPDs by regressing observed product-level GWP on the engineering-computed GWP for plants with available EPD data and applying the fitted regression to the remaining plants. For imputed observations and for subsequent counterfactual fuel scenarios, I assume a clinker-to-cement ratio of 0.914, consistent with the industry average for standard Portland cement (Type I/II). The resulting plant-level GWP values, computed under each plant's observed fuel mix and under counterfactual fuel scenarios, form the basis for evaluating Buy Clean eligibility in the counterfactual simulations.

Figure 12 presents the distribution of plant-level Global Warming Potential (GWP) computed using the engineering approach described above. The red dashed line denotes the Buy Clean emissions threshold of 922 kg CO_2 per ton of cement, which serves as the benchmark for determining policy eligibility. Among the 90 plant observations with sufficient data, 30 fall below the threshold while 60 exceed it. This distribution suggests that the Buy Clean standard is potentially binding for a substantial share of the industry.

6.2 Policy Implementation

I simulate six Buy Clean designs on the 2019 market. They differ in coverage, in the share of demand subject to the standard, and in the threshold. Designs (a) to (c) cover the 13 states of the Federal-State Buy Clean Partnership; designs (d) to (f) cover every county. The green share of demand in a covered county is $\lambda = 0.5$ or 0.6. The base threshold is 922 kg CO_2 per ton, the current benchmark, or 900. Two rules adjust the threshold by county, as in Section 3.3. In a covered county the threshold rises in steps of 10 kg/t until at least two plants within 300 miles are eligible; a county with fewer than two nearby plants is not covered. If an eligible plant is pushed above 1.2 times its capacity, the threshold rises in the counties where it sells to green buyers until one more plant qualifies. The 13-state designs cover 24.5 percent of national demand; the national designs cover 91.7 percent.

Fuel switching is part of every simulation. The switching cost takes the values of the first-stage confidence set at $\sigma = 15$ and 20, K between \$67 and \$82 million, and the private cost shock ν_j is drawn ten times for each plant. The same draws are used under every policy. For each (K, σ) point and draw, I solve the switching game by iterated best response: starting from the observed fuels, each coal plant switches if the gain in its owner's procurement profit over ten years, with the policy in place, covers $K + \nu_j$, and plants are revisited until no plant wants to change. This gives 40 equilibria per policy. I report the mean over the 40 runs and, where it matters, the range over the four (K, σ) points.

The baseline is no policy with the same switching game. Without any policy, 5.4 of the 66 coal plants switch on average, against 5 observed between 2019 and 2022. A plant is tipped by a policy if it is on gas under the policy and not under no policy in the same run. All changes below are against this baseline.

Equilibrium under the Counterfactual

Each county m is partitioned into a green submarket of mass $\lambda_m M_{mt}$ and a conventional submarket of mass $(1 - \lambda_m)M_{mt}$. The inclusive-value sums differ across submarkets because the choice sets do:

$$IV_{mt}^{\text{green}} = \sum_{k \in \mathcal{G}_m(\bar{e}_m)} \exp\left(\frac{\bar{u}_{kmt}}{1-\sigma}\right), \qquad IV_{mt}^{\text{conv}} = \sum_{k \in \mathcal{J}_m} \exp\left(\frac{\bar{u}_{kmt}}{1-\sigma}\right), \tag{31}$$

where $\mathcal{G}_m(\bar{e}_m) \subseteq \mathcal{J}_m$ is the set of plants eligible to serve green buyers in market m. Submarket-specific shares s_{jmt}^b, for $buy \in \{\text{green, conventional}\}$, follow from the nested-logit expression in equation (16) with the relevant IV_{mt}^b, where $s_{jmt}^{\text{green}} = 0$ for ineligible plants $j \notin \mathcal{G}_m$. Plant

j's total quantity aggregates demand from both buyer types across all served counties:

$$Q_{jt} = \sum_m M_{mt} \left[\lambda_m \, s_{jmt}^{\text{green}} + (1 - \lambda_m) \, s_{jmt}^{\text{conv}} \right]. \tag{32}$$

Because marginal cost $c_{jt}(Q_{jt})$ depends on this aggregated quantity, plants implicitly pool capacity across green and conventional segments. The joint equilibrium is solved as a single fixed point on the bid vector, $b_{jt}^* = c_{jt}(Q_{jt}(\mathbf{b}^*))$, by Gauss-Seidel coordinate bisection as in the baseline.

6.3 What Buy Clean Does

Table 2 summarizes the six designs. Figure 14 in Appendix A2 plots the switching, the emissions avoided and the change in intensity.

Table 2: Buy Clean with fuel switching: six policy designs, 2019 market

	Demand	Plants	CO_2 avoided, Mt		Domestic	Domestic		Welfare	Cost per
Design	covered	tipped	total	domestic	intensity	output	Price	ΔW_1, \$M	ton, \$
(a) 13 states, $\lambda = 0.5$, 922	24%	8.7 [7, 10]	2.0 (-2.2%)	+0.4	-1.9%	+1.4%	+1.5%	−169	85 [76, 96]
(b) 13 states, $\lambda = 0.6$, 922	24%	10.2 [9, 12]	2.3 (-2.5%)	+0.4	-2.1%	+1.7%	+1.9%	−185	82 [73, 92]
(c) 13 states, $\lambda = 0.5$, 900	24%	9.2 [8, 11]	2.0 (-2.2%)	+0.5	-1.9%	+1.4%	+1.5%	−177	89 [78, 104]
(d) national, $\lambda = 0.5$, 922	92%	9.0 [7, 11]	2.5 (-2.7%)	-1.5	-1.9%	+3.8%	+3.6%	−176	70 [64, 77]
(e) national, $\lambda = 0.6$, 922	92%	7.1 [6, 9]	2.1 (-2.3%)	-2.8	-1.2%	+4.7%	+4.5%	−155	75 [68, 81]
(f) national, $\lambda = 0.5$, 900	92%	19.0 [16, 22]	3.9 (-4.2%)	-0.2	-3.5%	+3.8%	+3.5%	−270	70 [63, 79]

Notes. Every entry is the mean over the 40 best-response runs (four (K, σ) points of the first-stage confidence set times ten draws of the private cost shock), measured against no policy with switching in the same run. Demand covered: share of national demand in covered counties after the two-plant rule. Plants tipped: coal plants on gas under the policy and not without it, with the 10th and 90th percentiles over runs in brackets; 5.4 plants switch without any policy. CO_2 avoided counts domestic kilns and imports at 880 kg/t; a negative domestic entry means domestic emissions rise as domestic output replaces imports. Price: quantity-weighted price of domestic cement. ΔW_1: change in buyer surplus plus producer surplus minus the annualized switching cost, million model dollars per year. Cost per ton: $-\Delta W_1$ per ton of CO_2 avoided, with the range over the four set points in brackets.

The current design, (a), tips 8.7 coal plants to gas, 10 percent of the industry's 91 plants and 13 percent of the plants that could switch. The range over the first-stage set is 7.6 to 9.8 plants, and the 10th to 90th percentiles over cost draws are 7 to 10. Raising the green share to 0.6 adds 1.5 plants; tightening the threshold to 900 adds 0.5. Going national at the same threshold adds almost nothing, (d) tips 9.0, because the capacity rule raises thresholds in 2,219 counties where eligible plants would otherwise be overloaded, which admits coal plants

without their switching. The national design with the 900 threshold, (f), tips 19 plants, 29 percent of the coal fleet. Coverage and stringency work together: a stringent threshold applied everywhere is what moves the most plants.

Emissions fall by 2.2 percent under (a) and by 2.2 to 2.7 percent under the other designs at 922, and by 4.2 percent under (f). Where the reduction comes from matters. Under the 13-state designs, 1.55 of the 2.0 Mt avoided are displaced imports; domestic kilns avoid 0.45 Mt, because the switchers' lower intensity is partly offset by higher domestic output. Under the national designs, domestic emissions rise: domestic output grows 3.8 to 4.7 percent as it replaces imports, which fall by 35 to 43 percent, and all of the reduction is at the border. Domestic emission intensity falls by 1.9 percent under (a) and (d) and by 3.5 percent under (f), the only design in which switching, rather than displaced imports, is the main source of abatement.

6.4 How the Policy Changes the Market

Table 9 reports output, marginal cost, price and markup for three groups of plants under designs (a) and (d): the 30 plants eligible on their current fuel, the plants the policy puts on gas, and the ineligible plants. Figure 16 shows the same changes in percent.

Reserving half of covered demand for eligible plants moves demand toward them. Eligible plants sell more: 1.5 percent for the incumbents and 5.0 percent for the switchers under (a). With convex costs, they move up their marginal cost curves: marginal cost rises 1.5 and 5.3 percent, and the median eligible plant's utilization goes from 0.82 to 0.87. They pass the higher cost on, so their prices rise by 1.3 and 5.3 percent. Ineligible plants lose sales, 0.7 percent, move down their cost curves, and bid slightly less. This is what a competitive model with capacity constraints would predict, and it is most of the price change: of the $1.3 per ton increase for eligible incumbents, $1.1 is cost. The national design is the same pattern at a larger scale, because demand is drawn away from imports rather than from ineligible plants: eligible plants' marginal cost rises 4.3 percent, and ineligible plants also sell more.

The part a competitive model cannot produce is in the last column. Markups rise for every group, including the ineligible plants whose sales fall. Eligible plants earn more because green buyers have fewer bidders: on sales to green buyers the markup is $27.3 per ton under (a), against $24.5 on their other sales and before the policy. Ineligible plants earn more for a different reason. The eligible plants they compete with for conventional buyers now bid higher, so the runner-up in those auctions is weaker, and the ineligible winner's markup rises by 1.2 percent, even as it sells less. Where the two-plant rule raised the threshold, plants

above 922 kg/t serve green buyers themselves, at a markup of $29.1 per ton, because in those counties they face even fewer rivals. The policy therefore creates market power for both the clean and the dirty suppliers at once. With switching, the sums are modest, a few hundredths of the price, because the plants the policy tips enter the eligible set and compete the rent away: before any plant responds, the same design raises eligible plants' markup on green sales to $44.5 per ton, 76 percent above the no-policy level, and the premium green buyers pay to $36 per ton (Table 11). That short-run rent is the incentive that tips the switchers; the equilibrium markup is what is left once they have switched.

6.5 Welfare

Table 3 reports who gains and who loses, in million dollars per year, for designs (a), (d) and (f) and for the carbon tax discussed next. Figure 19 plots the same.

Table 3: Who gains and who loses, with fuel switching, million dollars per year

	Buy Clean (a)	Buy Clean (d)	Buy Clean (f)	Tax $25	Tax $50	Tax $100
Green buyers	−212	−396	−435			
Conventional buyers	−56	−134	−118			
All buyers	−268	−530	−553	−1803	−3508	−6649
Plants already on gas	+25	+75	+81	−49	−97	−180
Switched plants, net of switching cost	+54	+81	+105	−53	−172	−524
Plants on coal	+15	+176	+71	−283	−465	−617
Other plants	+5	+22	+25	−14	−28	−54
Tax revenue	+0	+0	+0	+2175	+4081	+7143
ΔW_1, before climate benefits	−169	−176	−270	−27	−189	−881
ΔW_3, with avoided damages at the SCC	+154	+231	+355	+809	+1519	+2474
CO_2 avoided, Mt	2.0	2.5	3.9	5.2	10.6	20.8
Welfare cost per ton, domestic + imports	85	70	70	5	18	42
Welfare cost per domestic ton	435		4397	7	23	53

Notes. Changes against no policy with switching, means over the 40 runs, million model dollars per year. Buyer surplus is split between green and conventional buyers in proportion to the green share λ of each covered county; under the tax there are no green buyers. Switched plants are plants on gas under the policy, net of the annualized switching cost $(K+\nu_j)/7.72$ of the plants the policy tips. ΔW_3 values avoided CO_2, domestic and imported, at $162 per ton in model dollars ($190 in 2020 dollars). Welfare cost per ton: $-\Delta W_1$ divided by CO_2 avoided; the domestic denominator excludes displaced imports and is undefined where domestic emissions rise.

Buyers pay. Under (a), green buyers lose $212 million a year and conventional buyers $56 million. Most of it goes to plants: $25 million to plants already on gas, $54 million to the switchers net of their switching cost, $15 million to plants on coal. The net loss is $169 million a year, $85 per ton of CO_2 avoided, with a range of $76 to $96 over the first-stage

set. Valued at the social cost of carbon, $162 per ton in model dollars, the avoided emissions are worth $323 million, so the policy raises welfare by $154 million once climate benefits are counted. The national designs cost $70 to $75 per ton, and the stringent national design (f), which abates the most, costs $70.

Two things qualify these numbers. First, the cost per ton counts displaced imports as avoided emissions. Under the 13-state designs, they are three-quarters of the total. Per ton avoided at domestic kilns, the cost is $435 under (a), and under the national designs, domestic emissions do not fall. The policy is a cheap way to abate only if imported cement is counted. Second, with fuels fixed, the same design costs $257 per ton and lowers welfare even after climate benefits (Table 11). The investment response is what makes the policy pass a cost-benefit test, and a fixed-technology evaluation would reject it.

6.6 Comparison with a Carbon Tax

Table 4 reports the border-adjusted carbon tax at $25 to $200 per ton, with the same switching game. The tax enters every plant's bid at its own intensity, and imports pay at 880 kg/t, so no supplier is excluded and the supplier set in every county is unchanged.

Table 4: Border-adjusted carbon tax with fuel switching, 2019 market

	\$25	\$50	\$100	\$200
Plants tipped (of 66)	3.6	8.7	19.8	29.4
Domestic output, %	−4.2	−8.4	−17.1	−34.8
Imports, %	−10.5	−20.4	−37.9	−62.2
Price, \$/t	+19.9	+39.8	+79.8	+161.1
Industry markup, \$/t	−0.00	+0.01	+0.05	+0.34
Markup of clean plants ($\leq$ 922), \$/t	+0.03	+0.08	+0.20	+0.54
Markup of dirty plants, \$/t	−0.03	−0.08	−0.20	−0.27
Emissions, %	−5.6	−11.5	−22.5	−41.5
CO_2 avoided, Mt (domestic + imports)	5.2	10.6	20.8	38.3
CO_2 avoided from domestic kilns, Mt	4.0	8.3	16.5	31.2
Buyer surplus, \$M	−1803	−3508	−6649	−11922
Producer surplus, \$M	−369	−684	−1199	−1990
Switching cost, \$M	30	78	177	266
Tax revenue, \$M	2175	4081	7143	10784
ΔW_1, \$M	−27	−189	−881	−3394
ΔW_3, \$M	+809	+1519	+2474	+2791
Welfare cost per ton, \$	5	18	42	89

Notes. Tax in model dollars per ton of CO_2, levied on every ton of domestic emissions at each plant's own intensity and on imports at 880 kg/t. Changes against no policy with switching, means over the 40 runs. Clean plants are those at or below 922 kg/t after switching. The CO_2 avoided row counts domestic kilns and imports; the next row domestic kilns only.

A \$50 tax tips 8.7 coal plants, the same number as Buy Clean (a) (Figure 20). It cuts emissions by 11.5 percent, five times as much, because it reaches every margin: domestic output falls 8.4 percent, imports 20 percent, and the switchers lower their intensity. Markups do not move. The tax raises every plant's cost by nearly the same amount, so the gap between a winner's cost and the runner-up's is unchanged; clean plants' markup rises by 9 cents and dirty plants' falls by 7 cents (Figure 21). By lowering output it relaxes capacity constraints rather than tightening them: plants above capacity fall from 13 to 5.

Buyers lose far more under the tax, \$3.5 billion against \$0.3 billion, but \$4.1 billion returns as revenue, and every group of plants loses. The net welfare cost is \$189 million, \$18 per ton of CO_2 avoided, against \$85 under Buy Clean (a); per domestic ton it is \$23 against \$435. Figure 22 plots abatement against welfare for the tax grid and the six Buy Clean designs. Every Buy Clean point lies below the tax curve: at 2 to 4 Mt of abatement the tax curve lies at about \$10 to \$25 million of welfare loss, the Buy Clean designs at \$155 to \$270 million.

The difference is the design, not the carbon price. A tax attaches the same price to every ton at every supplier. A dirty plant keeps a sale when its cost advantage exceeds the carbon it adds, and abatement happens wherever it is cheaper than the tax. An eligibility rule prices nothing. It excludes rivals outright, so the rent it creates is set by the distance to the next eligible plant rather than by carbon saved, it is paid on every green ton whether or not the plant abated, and it concentrates demand on a few plants that move up their cost curves while cheaper plants and imports sit idle. The tax mitigates these distortions rather than avoiding them: it also lowers consumption in a market where prices already exceed marginal cost. But it does so at the carbon price, on every ton, and what buyers pay is collected as revenue rather than accruing to plants as scarcity rents.

7 Conclusion

This paper examines how green procurement policies affect firms' incentives to adopt cleaner production technologies. Using the U.S. cement industry and the Buy Clean Initiative as a setting, I develop a two-stage structural model that links fuel-switching investment decisions to downstream procurement competition. I show that green procurement operates not only by increasing demand for cleaner products, but also by reshaping competition through restrictions on supplier eligibility. In concentrated markets, these restrictions raise markups for eligible plants while also benefiting ineligible plants through equilibrium adjustments in production costs and competitive pressure. Under the current Buy Clean policy, average emissions intensity declines, but an increase in total cement output leads to higher aggregate emissions when fuel choices are held fixed. At the same time, the policy creates incentives for cleaner production by increasing the returns to fuel switching, although these incentives remain limited by the geographic coverage of green procurement demand and the substantial fixed costs of investment. Extending procurement standards nationally induces considerably more switching, while tightening emissions thresholds has relatively little effect. These findings highlight a central tradeoff in demand-side environmental policy: restricting procurement to cleaner suppliers can encourage investment in cleaner production, but may also weaken competition and generate unintended market responses. More broadly, the effectiveness of green procurement depends not only on environmental standards, but also on market structure, policy coverage, and firms' endogenous investment decisions.

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A Appendix

A1 Tables and Figures

A1.1 Summary Statistics: U.S. Cement Market

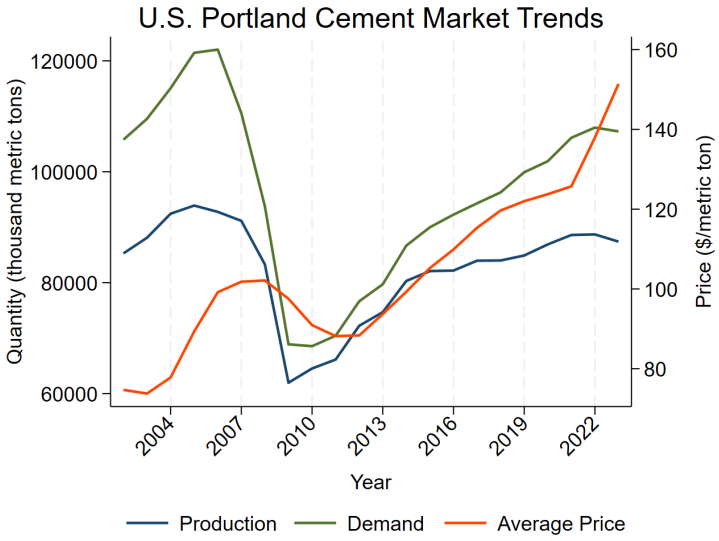


Figure 1: Demand, Production, and Price in the U.S. Cement Market

Notes: Annual U.S. Portland cement demand, domestic production, and national average price from 2002–2022. Data are from the U.S. Geological Survey (USGS) Cement Mineral Yearbooks.

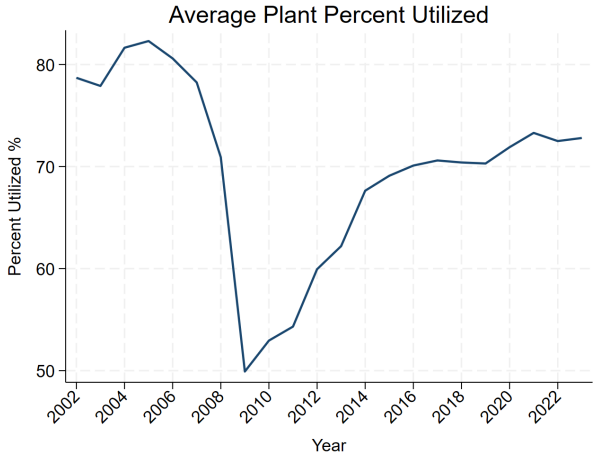
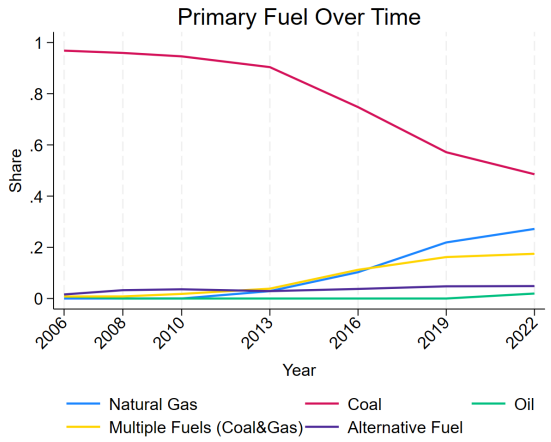


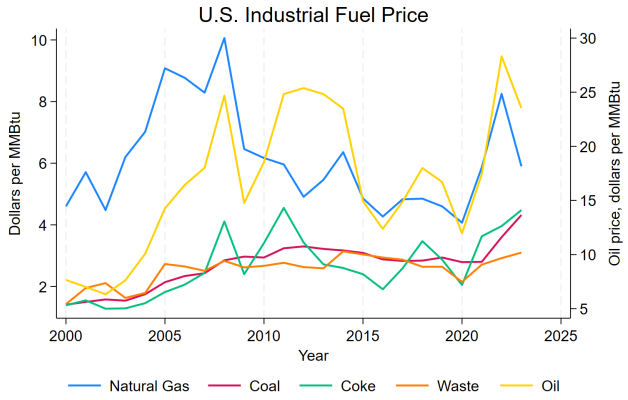
Figure 2: Average Plant Utilization Rate

Notes: Average annual plant utilization rate from 2002–2022 using USGS data. The decline in utilization rates following 2008 is consistent with the reduction in cement demand during the Great Recession.

A1.2 Fuel Transition and Fuel Prices



A: Fuel transition



B: U.S. delivery price in the industrial sector

Figure 3: Trends in Fuel Use and Fuel Prices

Notes: Panel A plots the share of U.S. cement plants by primary fuel type from 2006 to 2022. Observation years are irregular and reflect PIS publication frequency. Multiple Fuels (Coal&Gas) includes plants that list both coal or petroleum coke and natural gas as primary fuels. Alternative Fuel includes waste-derived materials such as tires, biomass, and industrial byproducts. Panel B plots U.S. average delivered fuel prices for the industrial sector in dollars per MMBtu, with oil prices shown on the right axis.

A1.3 The Decline of Coal Infrastructure

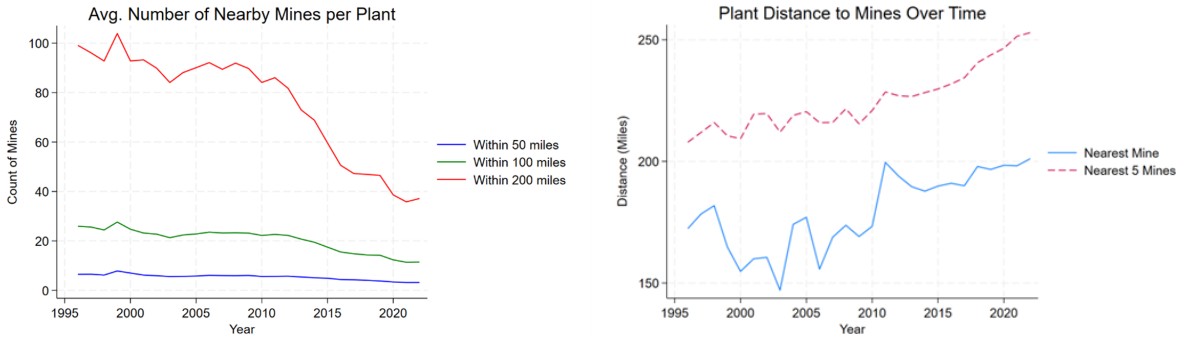


Figure 4: The Decline of Coal Mines

Notes: The left panel plots the average number of active coal mines within 50, 100, and 200 miles of each U.S. cement plant. The right panel plots the average distance from cement plants to the nearest active coal mine and to the average of the nearest five active mines.

A1.4 Natural Gas Pipeline Infrastructure

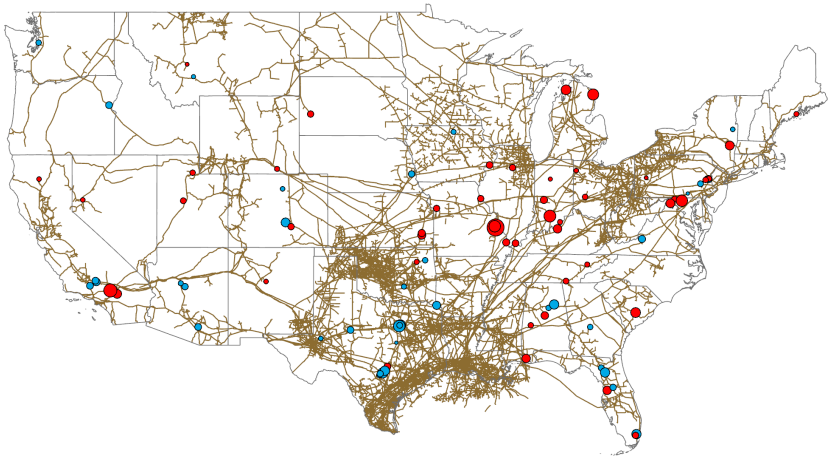
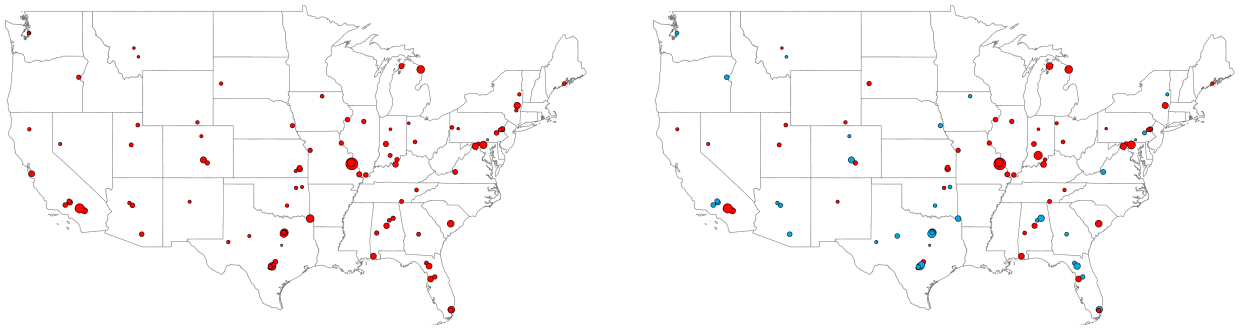


Figure 5: Natural Gas Infrastructure and the Geography of Fuel Switching

Notes: The figure shows the spatial distribution of U.S. cement plants and the national natural gas pipeline network in 2022. Red markers denote coal-fired plants, while blue markers denote natural-gas-fired plants. Marker size is proportional to plant production capacity. Brown lines represent major interstate natural gas pipelines.

A1.5 Spatial Distribution of Cement Plant Fuel Switching



Panel A: 2010 Panel B: 2022

Figure 6: Spatial Distribution of Cement Plant Fuel Use

Notes: The figures show the spatial distribution of U.S. cement plants and their primary fuel use in 2010 and 2022. Red markers denote coal-fired plants, while blue markers denote natural-gas-fired plants. Marker size is proportional to plant-level clinker production capacity.

A1.6 Cement Plant Fuel Switching by State

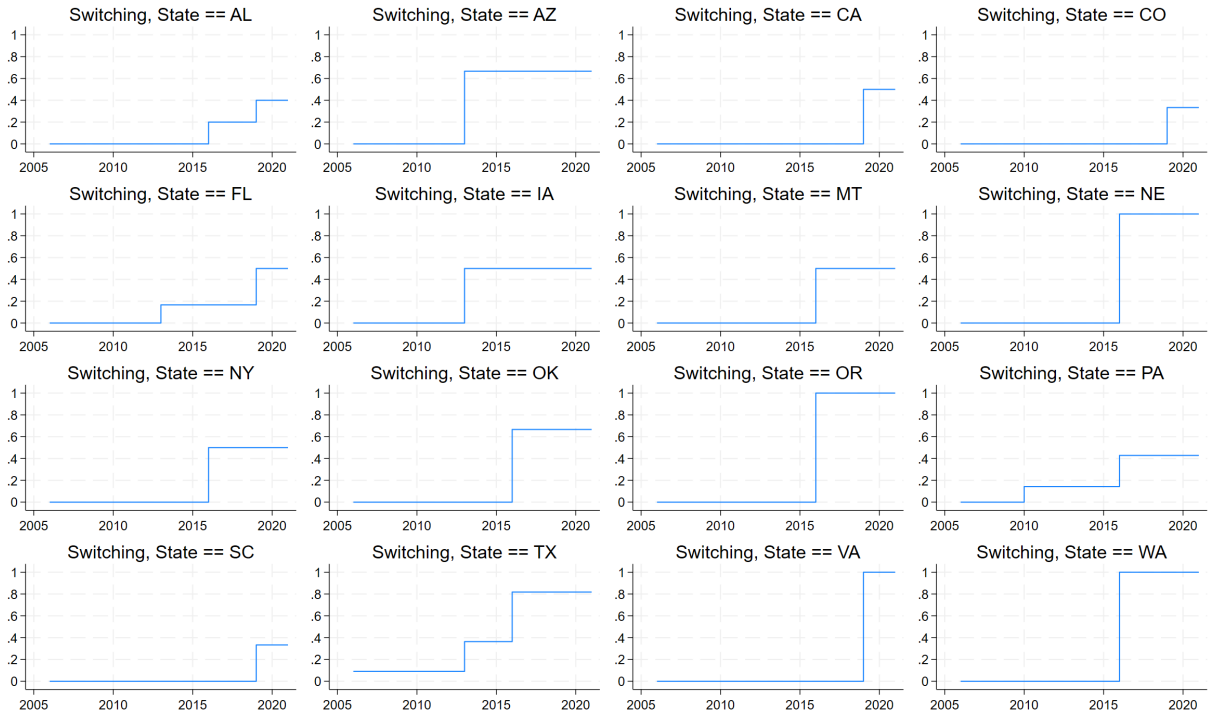
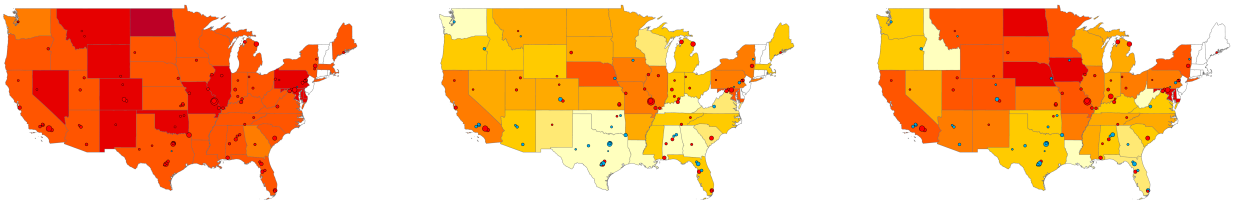


Figure 7: State-Level Cumulative Adoption of Natural Gas, 2006-2019

Notes: The figure plots the cumulative share of plants that have switched to natural gas by year, restricted to states with at least one switching plant during the sample period. Adoption in most states occurs in a single discrete period, though timing varies substantially across states. States such as Pennsylvania, Florida, Alabama, and Texas exhibit two waves of adoption within a relatively short time span.

A1.7 Spatial Distribution of Cement Plant Fuel Switching



Panel A: 2008 Panel B: 2019 Panel C: 2022

Figure 8: Spatial Variation in Energy Price: Coal/ Natural Gas Price Ratio

Notes: The figure shows the state-level coal-to-natural-gas price ratio in 2008, 2019, and 2022. Lighter colors indicate regions where coal prices are relatively higher compared to natural gas prices, implying stronger economic incentives for fuel switching toward natural gas.

Table 5: The Fuel Choice of Natural Gas

	(1) Gas as Primary	(2) Gas as Fuel	(3) Gas as Primary	(4) Gas as Fuel
Proximity to Gas Pipeline	−0.002	−0.014**	0.000	−0.018***
	(0.003)	(0.007)	(0.003)	(0.006)
Gas Price	−0.019*	−0.103***	−0.014	−0.108***
	(0.011)	(0.021)	(0.012)	(0.021)
Proximity to Pipeline × Gas Price	0.000	0.001	−0.000	0.002*
	(0.000)	(0.001)	(0.000)	(0.001)
Number of Coal Mines (500 Miles)	0.001	−0.002	−0.001**	−0.000
	(0.001)	(0.003)	(0.000)	(0.001)
Distance to Coal Mines			−0.000	0.001*
			(0.000)	(0.000)
Coal Price			0.002*	−0.002
			(0.001)	(0.002)
Kiln: Dry	−0.089*	0.094		
	(0.054)	(0.109)		
Kiln: Dry with Pro-heater	−0.062	0.076		
	(0.050)	(0.091)		
Kiln: Dry with Procalciner	−0.085*	0.132		
	(0.047)	(0.104)		
Constant	0.353***	1.456***	0.205*	1.500***
	(0.114)	(0.183)	(0.112)	(0.190)
Observations	636,000	636,000	534,000	534,000
Adj. R^2	0.222	0.143	0.224	0.153

Standard errors in parentheses
* p<0.10 ** p<0.05 *** p<0.01

A1.8 First Stage Estimation

Table 6: Switching cost estimates by estimation approach

	Probit, assumed information set			Moment inequalities	
	Perfect foresight	Current prices	Minimal information	Revealed preference	Dickstein–Morales
	(1)	(2)	(3)	(4)	(5)
Plant's expectation of the gain	realized V_{jt}	V_{jt}^{info}	$\mathbb{E}[V_{jt} \mid z_{jt}]$	unknown	unknown
Cost shock ν_{jt}	yes	yes	yes	no	yes
Switching cost K, \$ million	197 (59)	147 (35)	164 (56)	[7, 50]	[67, 82]
Cost shock σ, \$ million	120 (37)	94 (23)	104 (35)	0	[15, 20]
Annual equivalent of K, \$ million	25.5	19.0	21.2	[0.9, 6.4]	[8.7, 10.6]
Predicted switches in 2022 (5 observed)	5.6	7.4	6.9	[13, 40]	[5.6, 7.6]

Notes. Constant-cost model, definition 1: 441 plant-waves, 102 plants, 21 switches, waves 2008 to 2019. Columns (1) to (3): maximum likelihood of $P(\text{switch}) = \Phi((\bar{V} - K)/\sigma)$ with the stated measure of $\bar{V}$; standard errors clustered by plant in parentheses. Column (3) uses the fitted value of V on the twelve instruments (first-stage R^2 0.54). Column (4): the inequalities of Pakes et al. (2015), which are the Dickstein–Morales inequalities at $\sigma = 0$. Column (5): projections of the 95 percent confidence set for (K, σ) onto each parameter, over the values of σ that the 2022 holdout supports; the set at every σ is in Table 7. Annual equivalent: K divided by the 10-year annuity factor 7.72. Last row: the predicted number of switches among the 64 plants at risk in 2022, with the gain evaluated at the 2019–20 average prices; for the sets, the range over the accepted points.

Table 7: Common switching cost at each value of the cost shock, 95% sets

	K, \$ million	per year, \$ million	2022 predicted	Definition 2: K, \$ million
$\sigma = 0$	7 to 50	0.9 to 6.4	13 to 40	empty
$\sigma = 5$	50 to 74	6.4 to 9.6	5.0 to 13.1	20 to 37
$\sigma = 10$	55 to 74	7.1 to 9.6	5.4 to 10.9	27 to 41
$\sigma = 15$	67 to 74	8.7 to 9.6	6.1 to 7.6	37 to 45
$\sigma = 20$	74 to 82	9.6 to 10.6	5.6 to 6.9	empty
$\sigma = 30$	91	11.7	5.7	empty
$\sigma > 30$	empty			

Notes. Constant-cost model. Definition 1 in the first three columns (21 switches); definition 2, in which gas enters the set of primary fuels, in the last (422 plant-waves, 37 switches). The set is computed on a grid of $\log K$ in steps of 0.1, so adjacent bounds differ by about 10 percent. Under definition 2 the cost is lower because partial adoptions count as switches.

Table 8: Switching cost by group of plants, 95% sets, $\sigma = 15$

	Definition 1 (21 switches)		Definition 2 (37 switches)	
	Cost ratio e^{β}	K, \$ million	Cost ratio e^{β}	K, \$ million
All plants (constant cost)		67 to 74		37 to 45
No prior gas		≥ 67		37 to 100
Prior gas use in 2001–02	0.06 to 1.35	22 to 100	0.18 to 0.74	14 to 30
Near a pipeline (0.5 miles)		5 to 89		1 to 34
Far from a pipeline (8.6 miles); ratio per doubling	≥ 0.87	≥ 37	≥ 1.07	≥ 31
Decision before 2016		≥ 12		≥ 18
Decision from 2016 on	0.07 to 5.5	25 to 100	0.08 to 2.0	17 to 45
Reported costs, for comparison				
Keystone Cement, Bath PA, gas line (2019)		2.2		
EPA lateral cost, median plant / 8.6 miles		5 to 8 / 17 to 25		

Notes. Each shifter is estimated with a constant. Cost ratio: e^{β}, the factor by which the shifter multiplies the cost; for distance, 2^{β}, the factor per doubling of one plus the miles. K by group: $\exp(\beta' x)$ averaged over the group's plant-waves, range over the set. A one-sided entry means the other end lies on the edge of the grid. Near and far are the medians of the first and third distance terciles. Reported costs in model dollars: Keystone's line from the local distribution system (**globalcement2019**); EPA's lateral cost factor for an 8 to 12 inch line, \$1.9 to \$2.9 million per mile (**epa2024lateral**).

A1.9 Model Fit

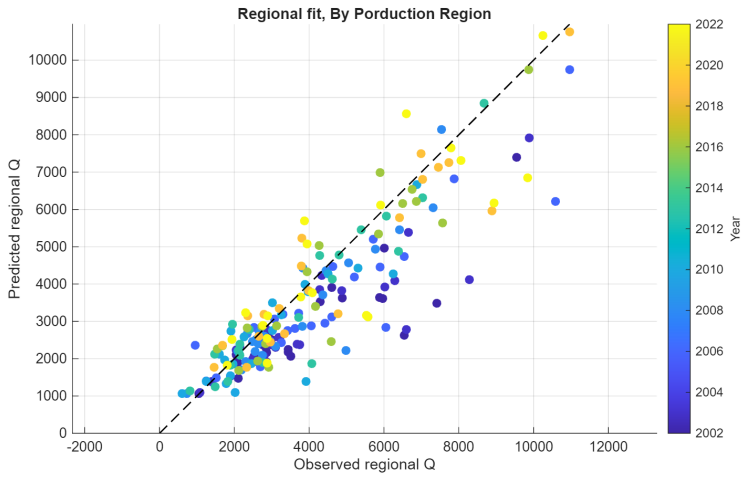


Figure 9: Model Fit: Predicted Quantity and Observed Quantity at Production Regions

Notes: The figure compares observed and predicted cement procurement quantities aggregated to the production-region level. Each point represents a production region-year observation. Predicted quantities are obtained from the estimated procurement model using the equilibrium market shares implied by the demand parameter estimates. The dashed line denotes the 45-degree line, along which predicted quantities equal observed quantities. Colors indicate calendar year.

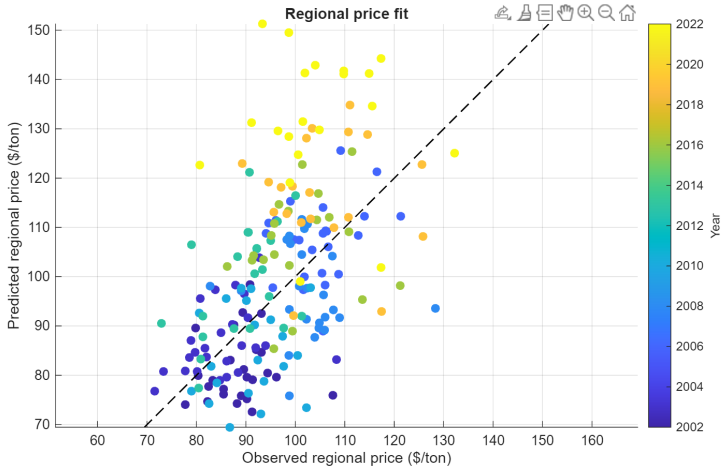


Figure 10: Model Fit: Predicted Average Price and Observed Average Price at Production Regions

*Notes:*The figure plots observed and predicted average delivered cement prices at the production-region level. Predicted prices are computed from the equilibrium procurement model using the estimated demand parameters, marginal cost function, and transportation costs. Each observation corresponds to a production region-year. The dashed line represents a perfect fit. Colors denote calendar year.

A1.10 Engineering-Computed and Observed Global Warming Potential

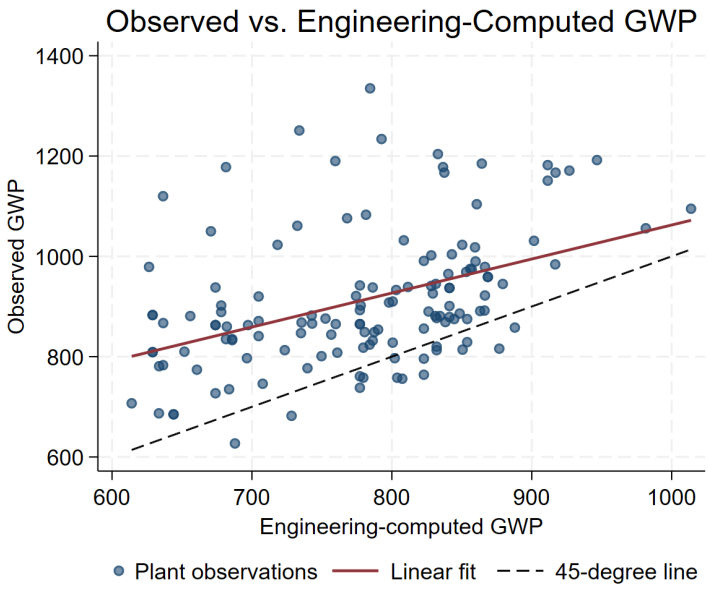


Figure 11: Engineering-Computed and Observed Global Warming Potential

Notes: Each point represents a plant-year observation. Engineering-computed GWP is constructed from combustion and process emissions associated with clinker production and adjusted by the clinker share in cement. The solid line reports the fitted linear relationship, and the dashed line represents the 45-degree line. Observed GWP is obtained from plant Environmental Product Declarations (EPDs).

A1.11 Global Warming Potential Distribution

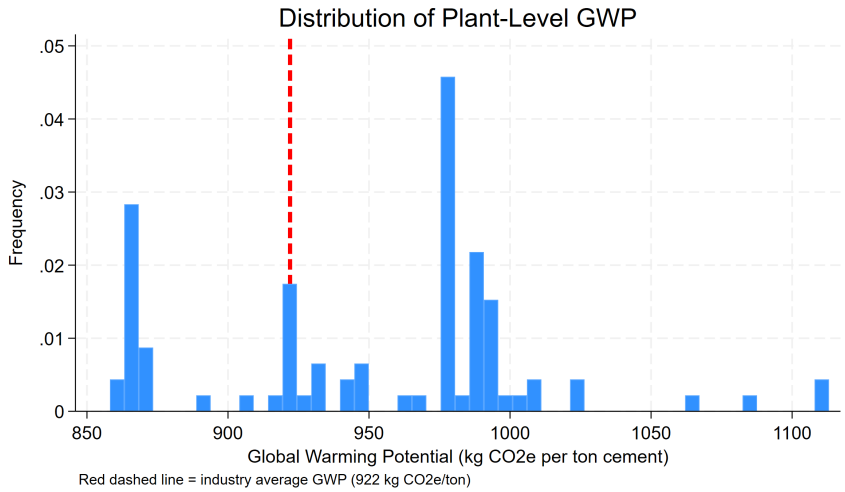


Figure 12: Observed Global Warming Potential Distribution

Notes: Plant-level engineering-computed GWP is constructed from combustion and process emissions associated with clinker production, assuming an industry-average clinker ratio of 91.4 %. The red dashed line indicates the Buy Clean benchmark of 922 kg CO_2 per ton of cement, which serves as the emissions threshold for policy eligibility.

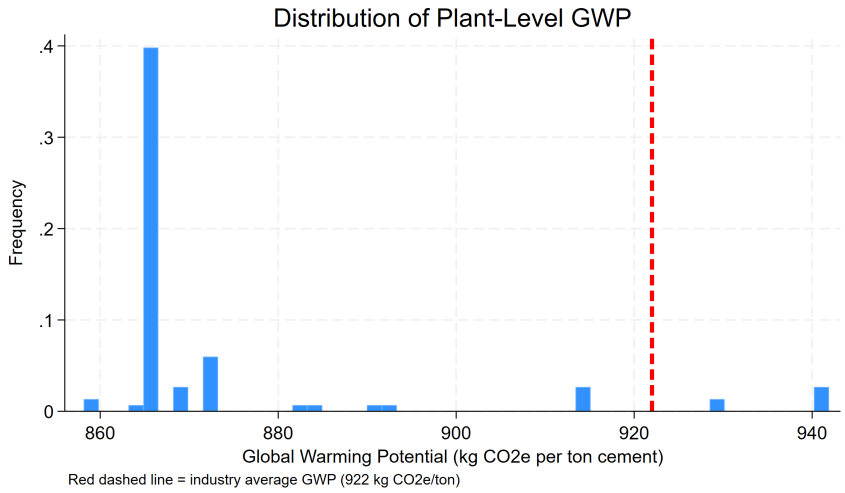


Figure 13: GWP Distribution in Counterfactual: If All Plants Switched to Gas

Notes: Plant-level engineering-computed GWP is assuming all plants switched to using natural gas. The red dashed line indicates the Buy Clean benchmark of 922 kg CO_2 per ton of cement, which serves as the emissions threshold for policy eligibility.

A2 Counterfactual Results: Details

All results in this appendix are with fuel switching and are measured against no policy with switching in the same run. A run is one (K, σ) point of the first-stage confidence set (K = 67, 74, 74 and 82 million dollars at σ = 15, 15, 20 and 20) and one draw of the private cost shock; there are 40 runs per policy, with the same draws under every policy. Means are over the 40 runs. Money is in million model dollars per year, emissions in Mt of CO_2 per year, prices in model dollars per ton.

Table 9: Plant groups under Buy Clean (a) and (d), with fuel switching

		Output, Mt	Marginal cost, $/t	Price, $/t	Markup, $/t
Buy Clean (a), 13 states					
Eligible on current fuel (30)	before → after	29.5 → 30.0	76.1 → 77.2	101.4 → 102.7	25.3 → 25.5
	change	+1.5%	+1.5%	+1.3%	+0.8%
Switchers (13)	before → after	19.6 → 20.6	71.5 → 75.3	94.7 → 99.7	23.2 → 24.4
	change	+5.0%	+5.3%	+5.3%	+5.1%
Ineligible (48)	before → after	37.0 → 36.8	80.6 → 80.3	105.7 → 105.7	25.1 → 25.4
	change	-0.7%	-0.3%	+0.1%	+1.2%
Buy Clean (d), national					
Eligible on current fuel (30)	before → after	29.5 → 30.7	76.1 → 79.4	101.4 → 105.0	25.3 → 25.6
	change	+3.9%	+4.4%	+3.6%	+1.3%
Switchers (14)	before → after	19.3 → 20.4	73.8 → 78.7	97.1 → 102.8	23.2 → 24.0
	change	+5.6%	+6.6%	+5.8%	+3.4%
Ineligible (47)	before → after	37.3 → 38.4	79.2 → 81.5	104.3 → 107.0	25.1 → 25.6
	change	+2.8%	+2.9%	+2.6%	+1.8%

Markup by buyer type, $/t (eligible at the base threshold / ineligible)

		No policy	Conventional buyers	Green buyers	Green sales, Mt
Buy Clean (a)	eligible	24.5	24.5	27.3	10.2
	ineligible	25.1	25.2	29.1	1.7
Buy Clean (d)	eligible	24.5	24.7	25.2	26.7
	ineligible	25.1	25.8	25.3	18.0

Notes. Means over the 40 runs; before is no policy with switching in the same run. Groups by status under the policy: eligible on current fuel (intensity at or below 922 kg/t without switching, 30 plants), switchers (on gas under the policy, intensity above 922 on their current fuel; 13 plants on average), ineligible (the rest). Output in million tons per year; marginal cost is the equilibrium bid; price and markup are quantity weighted over all sales. Lower panel: markups on sales to conventional and to green buyers under the policy, for plants at or below the base threshold and for the others; ineligible plants sell to green buyers only in counties where the two-plant or capacity rule raised the threshold.

A2.1 The Six Buy Clean Designs

Table 10: Buy Clean designs with fuel switching: market outcomes

	13 states 0.5, 922 (a)	13 states 0.6, 922 (b)	13 states 0.5, 900 (c)	national 0.5, 922 (d)	national 0.6, 922 (e)	national 0.5, 900 (f)
Demand covered, %	24.5	24.5	24.5	91.7	91.7	91.7
Counties raised by the two-plant rule	225	225	341	1033	1033	1454
Counties raised by the capacity rule	0	0	0	2219	2229	2260
Plants on gas, no policy	5.4	5.4	5.4	5.4	5.4	5.4
Plants on gas, policy	14.1	15.6	14.6	14.4	12.5	24.4
Plants tipped	8.7	10.2	9.2	9.0	7.1	19.0
range over set points	7.6 to 9.8	9.0 to 11.5	7.9 to 10.4	7.8 to 10.4	6.1 to 8.2	16.7 to 20.8
Eligible plants after switching	43	45	34	44	41	42
Domestic output, %	+1.4	+1.7	+1.4	+3.8	+4.7	+3.8
Imports, %	−13.7	−16.8	−13.7	−35.4	−43.5	−35.9
Price, $/t	+1.56	+1.89	+1.54	+3.69	+4.53	+3.56
Industry markup, $/t	+0.46	+0.50	+0.48	+0.48	+0.47	+0.56
Eligible: marginal cost, $/t	+2.2	+2.6	+2.3	+3.9	+4.7	+4.3
Eligible: markup, $/t	+0.59	+0.66	+0.61	+0.51	+0.57	+0.51
Eligible: markup on green sales, $/t	27.3	27.2	27.2	25.2	25.2	25.0
Ineligible: markup, $/t	+0.30	+0.28	+0.39	+0.45	+0.36	+0.65
Ineligible: output, %	−0.7	−0.5	−0.1	+2.8	+4.0	+1.6
Green minus conventional price, $/t	+1.5	+1.3	+1.7	−0.3	−0.8	−0.0
Plants above capacity (13.1 before)	15.1	15.5	15.0	16.4	19.2	16.4
Emissions, %	−2.2	−2.5	−2.2	−2.7	−2.3	−4.2
CO_2 avoided, Mt	2.0	2.3	2.0	2.5	2.1	3.9
from domestic kilns	+0.4	+0.4	+0.5	−1.5	−2.8	−0.2
from imports	1.6	1.9	1.6	4.0	4.9	4.1
Domestic emission intensity, %	−1.9	−2.1	−1.9	−1.9	−1.2	−3.5
Green buyers, $M	−212	−240	−219	−396	−480	−435
Conventional buyers, $M	−56	−66	−53	−134	−141	−118
Producer surplus, $M	+178	+214	+176	+436	+532	+458
Switching cost, $M	78	94	81	82	66	176
ΔW_1, $M	−169	−185	−177	−176	−155	−270
range over set points	−177 to −159	−197 to −174	−186 to −167	−184 to −167	−162 to −148	−290 to −251
ΔW_3, $M	+154	+183	+149	+231	+183	+355
Welfare cost per ton, $	85	82	89	70	75	70
Buyers' cost per ton, $	136	135	137	212	299	144

Notes. Means over the 40 runs against no policy with switching in the same run, 2019 market, mid parameter set. Designs: coverage (13 partnership states or national), green share λ of demand in covered counties, base threshold in kg CO_2 per ton. Thresholds: in a covered county the threshold rises in steps of 10 kg/t until two plants within 300 miles are eligible (two-plant rule); an eligible plant pushed above 1.2 times capacity raises the threshold where it sells green until one more plant qualifies (capacity rule); 141 (13 states) and 422 (national) targeted counties with fewer than two nearby plants are not covered. Eligible plants are plants at or below the base threshold after switching. Green price and conventional price are quantity-weighted over all plants. Buyers' cost per ton is $-\Delta CS$ per ton avoided.

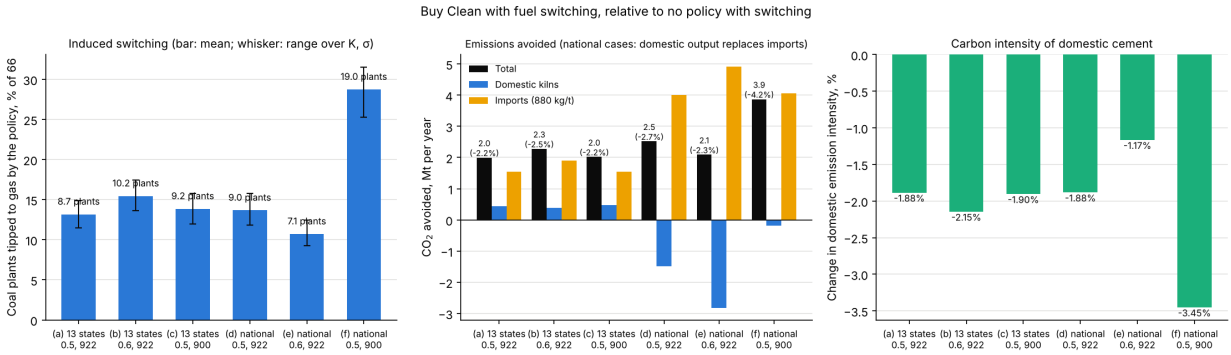


Figure 14: Buy Clean with fuel switching: induced switching, emissions avoided and carbon intensity, six designs

Notes. Left: coal plants tipped to gas by the policy as a share of the 66 coal, coke and oil plants; bar is the mean over the 40 runs, whisker the range over the four (K, σ) points. Middle: CO_2 avoided per year against no policy with switching, in total, at domestic kilns and at the border (imports at 880 kg/t); under the national designs domestic emissions rise because domestic output replaces imports. Right: change in the emission intensity of domestic cement. Labels: coverage, green share λ, base threshold.

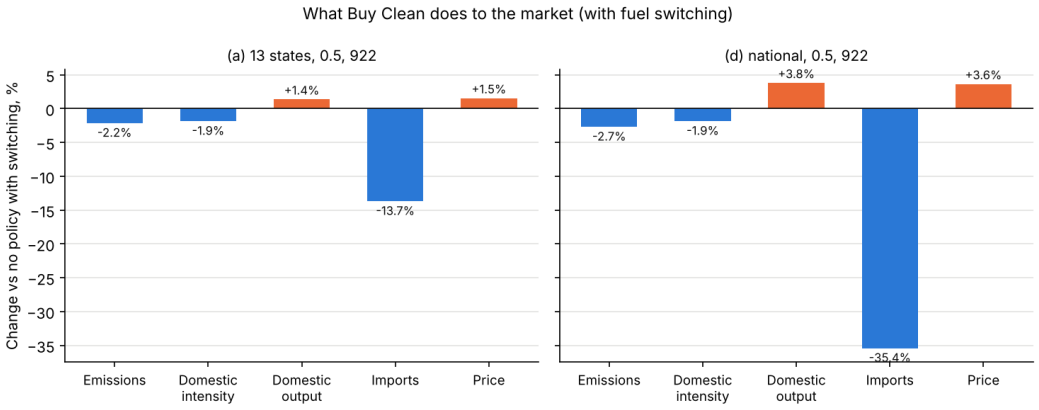


Figure 15: Market aggregates under Buy Clean (a) and (d), with fuel switching

Notes. Percent changes against no policy with switching, means over the 40 runs. Price is the quantity-weighted price of domestic cement. Under the national design domestic output rises as imports fall by a third, so total emissions fall while domestic emissions rise.

A2.2 Plant Groups and Buyer Types

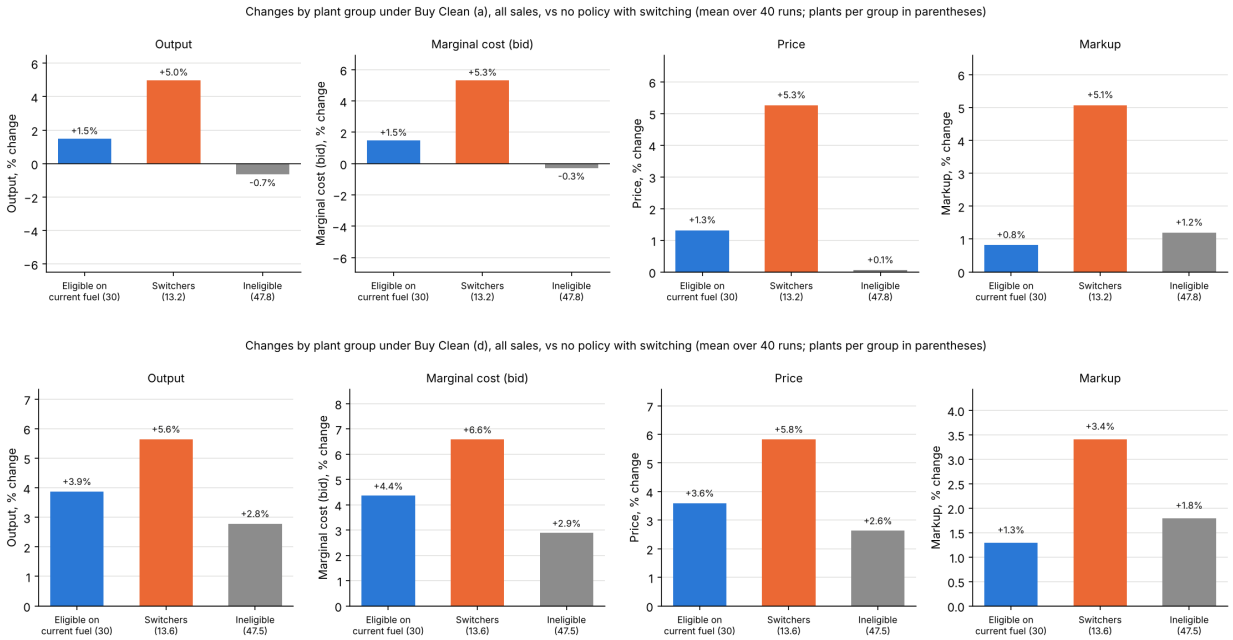


Figure 16: Changes by plant group under Buy Clean (a) (top) and (d) (bottom), all sales, with fuel switching

Notes. Percent changes against no policy with switching, means over the 40 runs. Groups by status under the policy: eligible on current fuel (intensity at or below 922 kg/t without switching), switchers (on gas under the policy and above 922 on their current fuel), ineligible (the rest); plants per group in parentheses, non-integer for the switchers because the count varies across runs. Marginal cost is the equilibrium bid; price and markup are quantity weighted over all of the group's sales. The first three panels are the cost pass-through a competitive model would produce; the markup panel is the part that requires endogenous markups: every group's markup rises, including the ineligible plants, whose sales fall.

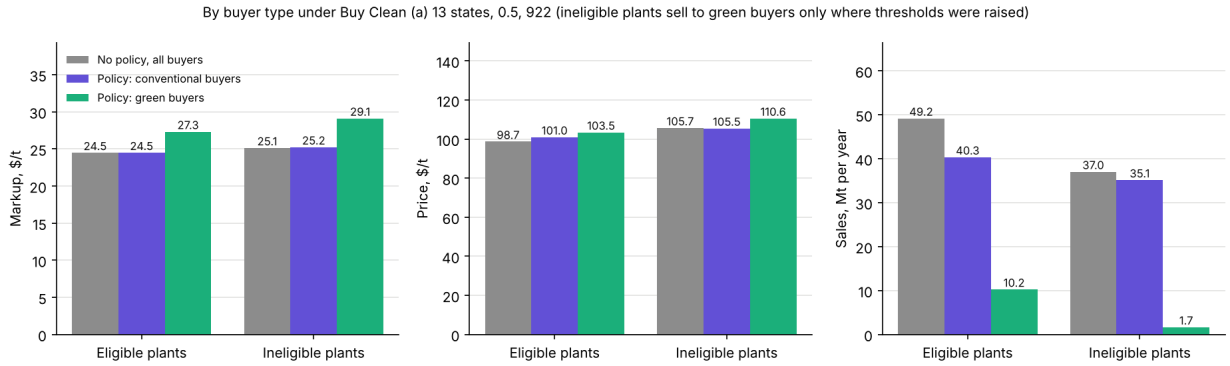


Figure 17: Markup, price and sales by buyer type under Buy Clean (a), with fuel switching

Notes. Plants at or below the base threshold of 922 kg/t after switching (eligible) and above it (ineligible). Gray: no policy with switching, all buyers. Under the policy, sales to conventional buyers and to green buyers are shown separately. Ineligible plants sell to green buyers only in counties where the two-plant or capacity rule raised the threshold above 922; in those counties they face few rivals, which is why their markup on green sales is the highest in the figure. Means over the 40 runs.

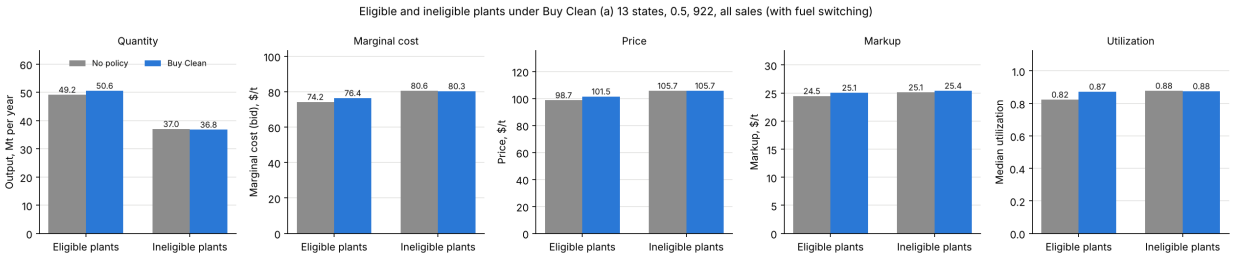


Figure 18: Eligible and ineligible plants under Buy Clean (a), levels, with fuel switching

Notes. Levels before (no policy with switching) and under the policy, quantity weighted over all sales; utilization is the median across the group's plants. Means over the 40 runs. The same figure for design (d) is available on request.

A2.3 Fuels Fixed and With Switching

Table 11: Why the investment response matters: Buy Clean with fuels fixed and with switching

	Buy Clean (a), 13 states		Buy Clean (d), national	
	fuels fixed	with switching	fuels fixed	with switching
Eligible plants	30	43	30	44
Markup on green sales, $/t (no policy 25.3)	44.5	27.3	27.1	25.2
Eligible plants' marginal cost, $/t	76.6 → 83.5	74.2 → 76.4	76.6 → 82.2	75.2 → 79.2
Green price minus conventional price, $/t	36	1.5	3	-0.3
Emissions, %	-1.5	-2.2	-1.6	-2.7
Buyer surplus, $M	-716	-268	-765	-530
Producer surplus, $M	+357	+178	+478	+436
ΔW_1, $M	-359	-169	-286	-176
Welfare cost per ton, $	257	85	195	70

Notes. Fuels fixed: every plant keeps its 2019 fuel, measured against no policy with fuels fixed. With switching: means over the 40 runs against no policy with switching. The fuels-fixed columns are the short-run effect, before any plant responds, and the incentive the policy creates; the switching columns are the equilibrium once the plants it tips have switched and compete the rent away.

A2.4 Welfare and the Carbon Tax

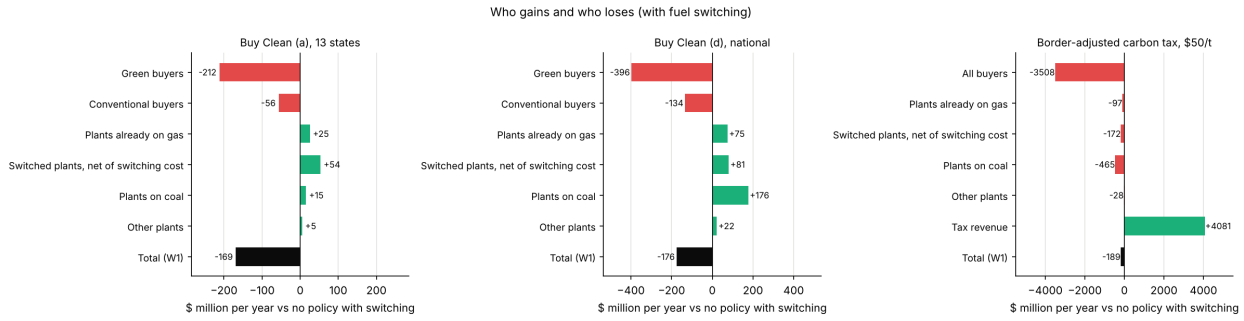


Figure 19: Who gains and who loses: Buy Clean (a), Buy Clean (d) and a $50 border-adjusted carbon tax, with fuel switching

Notes. Changes against no policy with switching, million model dollars per year, means over the 40 runs. Buyer surplus is split between green and conventional buyers in proportion to the green share of each covered county. Switched plants are net of the annualized switching cost of the plants the policy tips. Total is ΔW_1: buyer surplus plus producer surplus minus switching cost plus tax revenue, before climate benefits. Under Buy Clean the buyers' loss accrues mostly to plants; under the tax every plant group loses and the buyers' payment is collected as revenue.

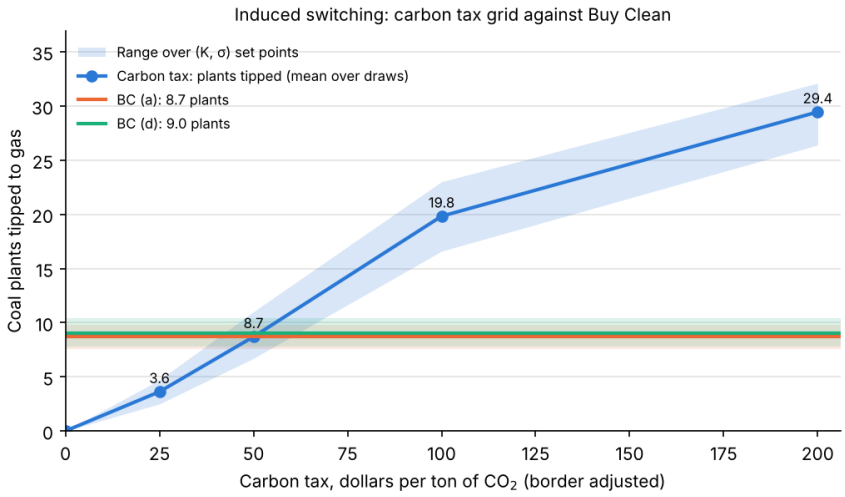


Figure 20: Induced switching under the carbon tax and under Buy Clean

Notes. Coal plants tipped to gas, against no policy with switching in the same run. Line: mean over the 40 runs at each tax rate; band: range over the four (K, σ) points of the first-stage set. Horizontal lines: Buy Clean (a) and (d), with their ranges. A \$50 tax tips the same number of plants as Buy Clean (a).

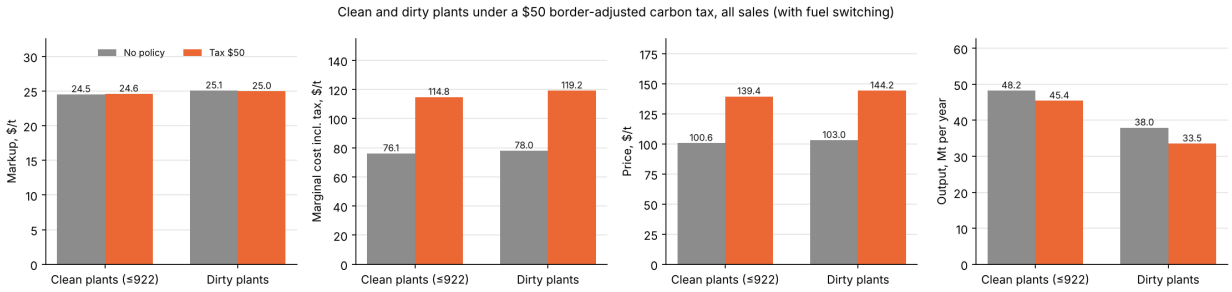


Figure 21: Clean and dirty plants under a \$50 border-adjusted carbon tax, with fuel switching

Notes. Plants at or below 922 kg/t after switching (clean) and above (dirty). Marginal cost includes the tax. The tax raises every plant's cost by nearly the same amount, so markups are unchanged while costs and prices rise by about \$40 per ton and output falls. Means over the 40 runs.

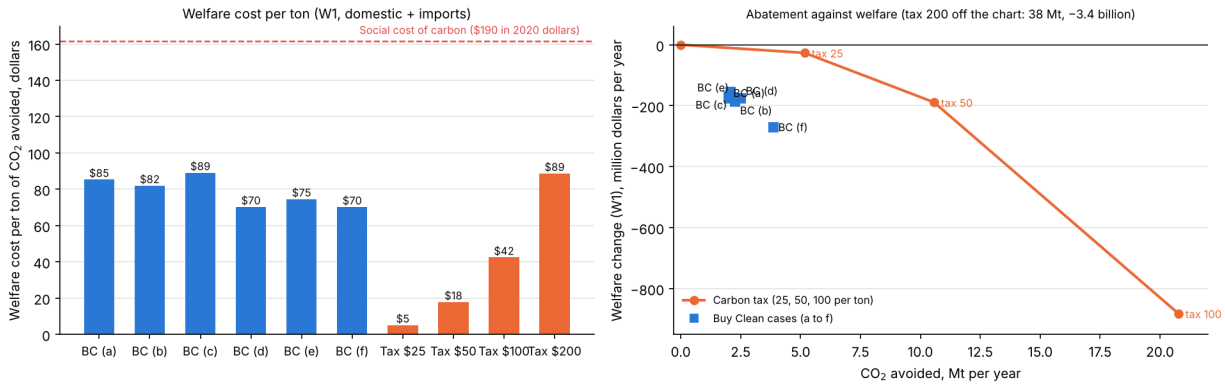


Figure 22: Welfare cost per ton of CO_2 avoided, and abatement against welfare

Notes. Left: $-\Delta W_1$ per ton of CO_2 avoided (domestic kilns plus imports), against the social cost of carbon of \$162 per ton in model dollars (\$190 in 2020 dollars). Right: CO_2 avoided against ΔW_1 for the tax grid (\$25, \$50, \$100; the \$200 point, 38 Mt at −\$3.4 billion, is off the chart) and the six Buy Clean designs. Every Buy Clean design lies below the tax curve at the same abatement.

A3 Data Construction

A3.1 Cost Shifter: Fuel Costs and Energy Requirements

I construct kiln-level fuel costs as the product of the fossil fuel price and the kiln's energy requirement:

$$\text{Kiln Fuel Cost}_{jt} = P_{jt}^{fuel} \times E_{jt}, \tag{33}$$

where P_{jt}^{fuel} is measured in dollars per MMBtu and E_{jt} is measured in MMBtu per metric tonne of clinker.

Fuel prices come from the State Energy Data System (SEDS). I use state-level average industrial-sector prices for coal, natural gas, and distillate fuel oil. I map each kiln's reported primary fossil fuel to the corresponding state-year fuel price. When a kiln reports multiple primary fossil fuels, the fuel shares are not observed. In the baseline specification, I assign P_{jt}^{fuel} as the simple average of the state-year prices of all listed primary fossil fuels.

As a robustness check, I also use a priority rule. If coal is listed among the primary fuels, I assign the coal price. If coal is not listed but natural gas is listed, I assign the natural gas price. Distillate fuel oil is assigned only when oil is the only listed fossil fuel. This rule is conservative because coal remains the dominant kiln fuel for much of the sample period.

Energy requirements are assigned by kiln technology. I use the energy requirements reported in Miller et al. Specifically, energy requirements are 3.94 for precalciner kilns, 4.11 for preheater kilns, 5.28 for long dry kilns, and 6.07 for wet kilns, all measured in MMBtu per metric tonne of clinker.

A3.2 Selection of Import Districts

This section describes the procedure for constructing the set of import districts included in the analysis. Imports are observed at the U.S. customs district level based on USGS data availability. I treat each customs district as a distinct import supplier. I exclude non-continental districts (Honolulu, HI, and San Juan, PR) to focus the analysis on mainland cement transactions and import ports.

To focus on economically constant import locations, I select import districts based on their contribution to aggregate import volume, using USGS data from study sample years (2002, 2003, 2006, 2008, 2010, 2013, 2016, 2019, and 2022). Specifically, I compute total import quantities for each district across the sample years and rank districts in descending order. I retain the smallest set of districts whose cumulative import share reaches 90 percent of total

imports in the sample.

To avoid including locations with sporadic or negligible activity, we further require selected districts to exhibit sustained import activity over time. In particular, districts must have positive (or above-threshold) import volumes in a substantial fraction of sample years. This ensures that the selected set captures stable import locations rather than transitory or idiosyncratic observations.

The resulting set of import districts represents the primary import entry points for cement in the United States over the sample period. All remaining districts are excluded from the analysis. In robustness checks, we verify that alternative selection rules (e.g., using all years or varying the coverage threshold) yield similar results.

A3.3 Market Size

To construct county-level cement demand, I combine national Input–Output (IO) coefficients from the Bureau of Economic Analysis (BEA) with county-level economic activity. Following a shift-share approach, national cement requirements provide predetermined industry shares, while local GDP provides geographic variation in demand.

Cement Requirements Let $\theta_{g,t}$ denote the dollar value of cement inputs required to deliver one dollar of final demand in industry g. I obtain these coefficients from the BEA industry-by-commodity Total Requirements Tables and aggregate them from the 6-digit to the 4-digit NAICS level. Because detailed IO tables are available only in benchmark years, I linearly interpolate coefficients between benchmark years:

$$\theta_{g,t} = \theta_{g,b_0} + \frac{t - b_0}{b_1 - b_0} \left(\theta_{g,b_1} - \theta_{g,b_0}\right).$$

Industry Aggregation I retain construction (NAICS 23) and mining (NAICS 21), which account for the majority of cement consumption and whose value added provides a reasonable proxy for final demand. Manufacturing industries are excluded because their output is primarily used as intermediate inputs, creating potential double-counting.

Using annual national value added data, I compute within-sector shares

$$s_{g,t} = \frac{VA_{g,t}}{\sum_{g \in G} VA_{g,t}},$$

and aggregate cement requirements to the 2-digit industry level:

$$\theta_{G,t} = \sum_{g \in G} s_{g,t} \theta_{g,t}.$$

County-Level Demand I use BEA county GDP by industry to measure local economic activity. County-level cement demand is

$$D_{c,t} = \sum_{G} \theta_{G,t} GDP_{c,G,t}.$$

To ensure consistency with observed cement consumption, I allocate USGS state-level cement demand across counties according to predicted demand shares:

$$Q_{c,t} = \frac{D_{c,t}}{\sum_{c \in s} D_{c,t}} Q_{s,t},$$

where $Q_{s,t}$ denotes total cement consumption in state s. This procedure preserves observed state-level totals while allowing county-level variation driven by differences in economic activity and industry composition.

A3.4 Calibration of Transportation Mode Choice and Travel Costs

This appendix describes the calibration of transportation mode choice and travel costs between production locations and buyer markets.

A3.4.1 Two-Mode Transportation Model

According to the USGS, in 2017, 46.6% of plant-to-terminal cement shipments occurred by rail, 36.4% by barge or boat, and the remainder by truck, while 98.7% of terminal/plant-to-customer shipments used trucking. The Commodity Flow Survey (CFS) similarly shows trucking dominates observed shipments. Since spatial data from the Bureau of Transportation Statistics indicate that nearly all cement plants have direct rail access, the relevant source of spatial variation in transportation costs arises from access to navigable waterways.

I therefore adopt a simplified two-mode representation: (i) land transport (combining truck and rail) and (ii) water transport. Calibration proceeds in three steps: assigning each CFS area to its nearest cement port or dock, computing waterway distances between docks, and

estimating the probability of water-based shipment as a function of shipment distance and waterway access.

A3.4.2 Transportation Cost Functions

Let i and j denote origin and destination CFS areas. Overland cost is linear in distance:

$$C_{ij}^{land} = \beta_L d_{ij}.$$

Water cost combines overland access to and from ports with the waterway segment:

$$C_{ij}^{water} = \alpha_W + \beta_L \left(d_{i,p(i)} + d_{p(j),j} \right) + \beta_W d_{p(i),p(j)}^{water},$$

where $p(i)$ and $p(j)$ denote the nearest ports and $\beta_W < \beta_L$. Shipments occur through the cheaper mode:

$$C_{ij} = \min \left\{ C_{ij}^{land},\ C_{ij}^{water} \right\}.$$

A3.4.3 Overland Distance

I use the CFS variable *distance_routed*, which records the total routed distance traveled along the transportation network. The origin to destination measure is a survey-weighted average of shipment-level routed distances:

$$d_{ij}^{land} = \frac{\sum_{s \in (i,j)} w_s \, d_s^{routed}}{\sum_{s \in (i,j)} w_s},$$

where w_s is the CFS survey weight. As a robustness check, I reweight by the product of survey weight and shipment tonnage q_s; results are qualitatively similar.

A3.4.4 Waterway Access and Distance

Regions follow the 2017 CFS metropolitan areas. Because these polygons are irregular, I generate 100 random points within each region following Donaldson and Hornbeck (2016) and compute the Euclidean distance from each point to the nearest dock. I snap dock locations to the nearest waterway segment within a 500-meter tolerance.

The waterway network is built from the National Waterway Network Links and Dock data from Waterborne Commerce Statistics Center (WCSC). I use Waterborne Cargo and Trips Data Files and Navigation Facilities to select all docks and ports with regular cement receipt

or shipment flows. Shortest-path distances between port pairs are computed using the Origin to Destination Cost Matrix solver.

Total water-based shipping distance combines overland access and the waterway segment:

$$d_{ij}^{water} = d_{i,p(i)} + d_{p(i),p(j)}^{water} + d_{p(j),j}.$$

A3.4.5 Mode Choice Calibration

Shipment distances from the CFS are grouped into bins, and within each bin I compute the empirical share of water shipments. I first verify with a linear probability model that water usage increases with proximity to waterways and with shipment distance. I then calibrate the cost parameters by matching predicted mode shares to the empirical shares across distance bins.

A3.4.6 Limitations

Several simplifications are involved. Transportation costs are proxied by distance rather than observed freight rates, which are unavailable for the cement industry and likely endogenous to local conditions. Costs are assumed symmetric across directions, which is reasonable for bulk shipments of homogeneous goods. The model abstracts from congestion, scale economies, and operational constraints that cause actual routes to deviate from shortest paths. Time costs are ignored since cement is a storable bulk commodity. The resulting measures should be interpreted as reduced-form proxies for the relative cost advantage of alternative modes across origin-to-destination pairs rather than exact shipment-level costs.